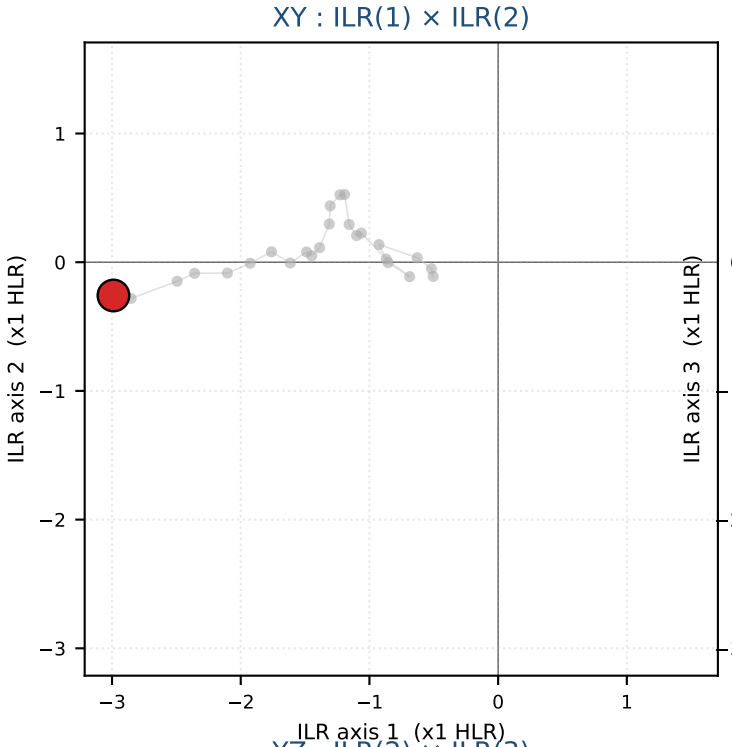
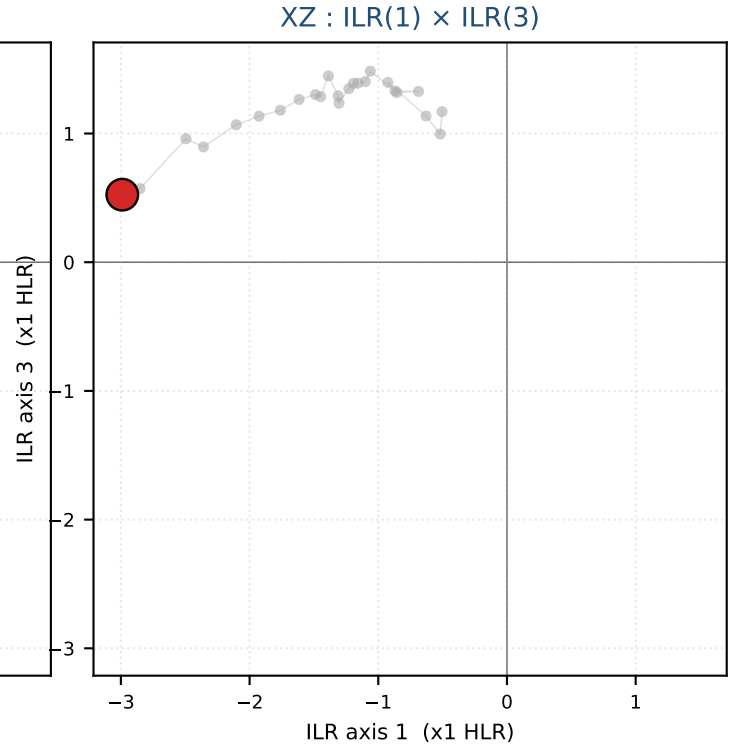
STAGE 1 v4 | t=0 | 2000 | D=9 T=26 | level 1 first-principles

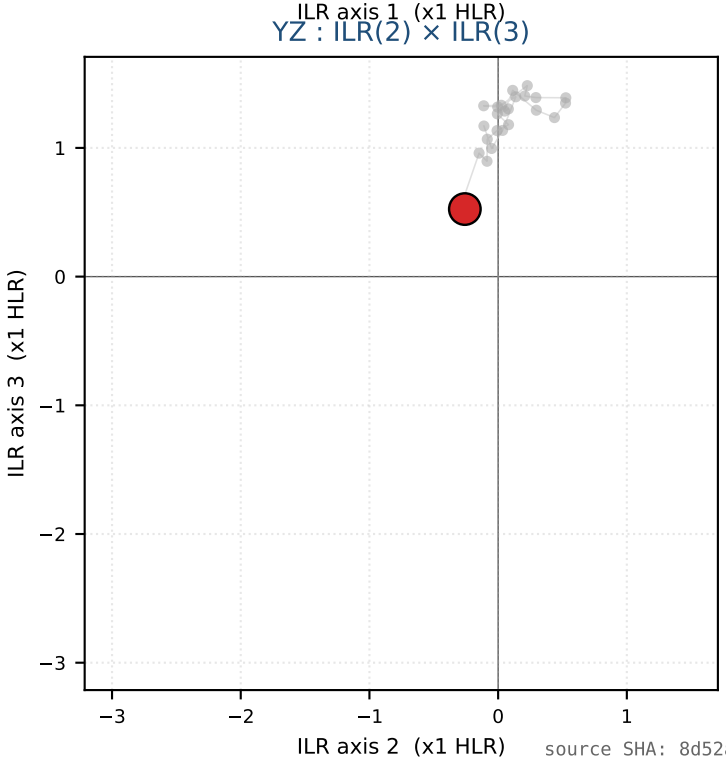
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

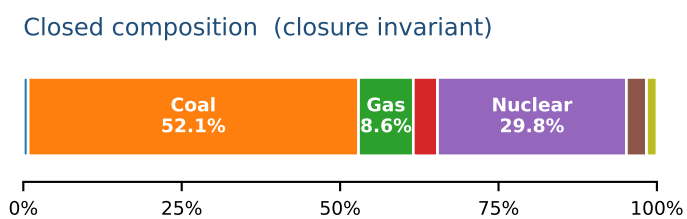




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0076	+6.430
Coal	0.5215	+10.657
Gas	0.0865	+8.861
Hydro	0.0382	+8.043
Nuclear	0.2981	+10.098
Other Fossil	0.0317	+7.858
Other Renewabl	0.0000	-29.574
Solar	0.0000	-29.574
Wind	0.0164	+7.200





closure: Σ = 1.000000 OK

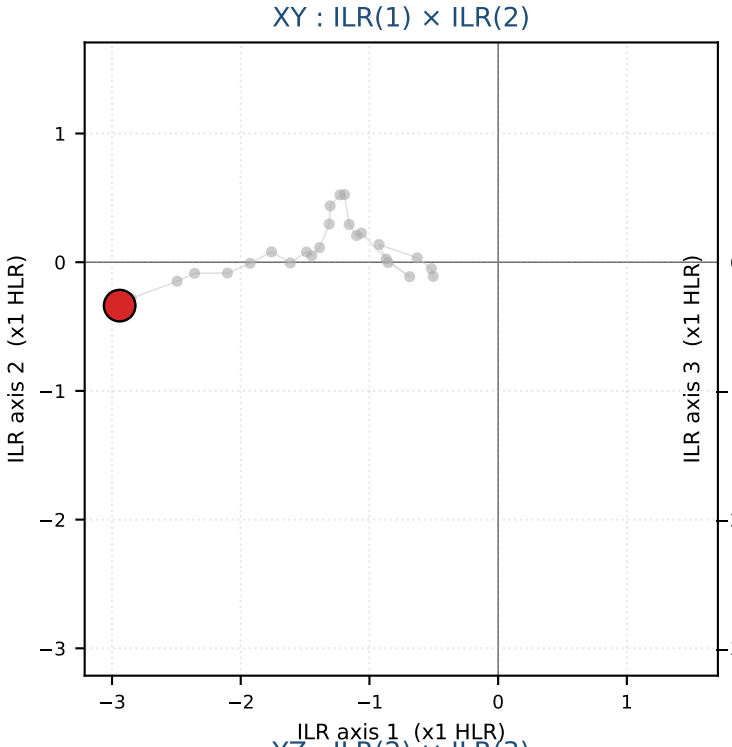
display factor: x1 (raw HLR x 1)

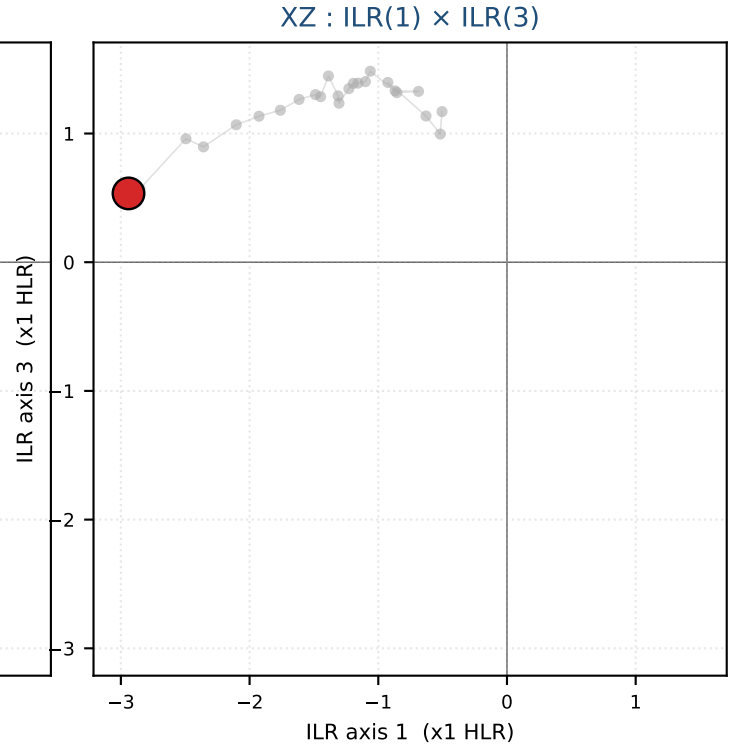
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=1 | 2001 | D=9 T=26 | level 1 first-principles

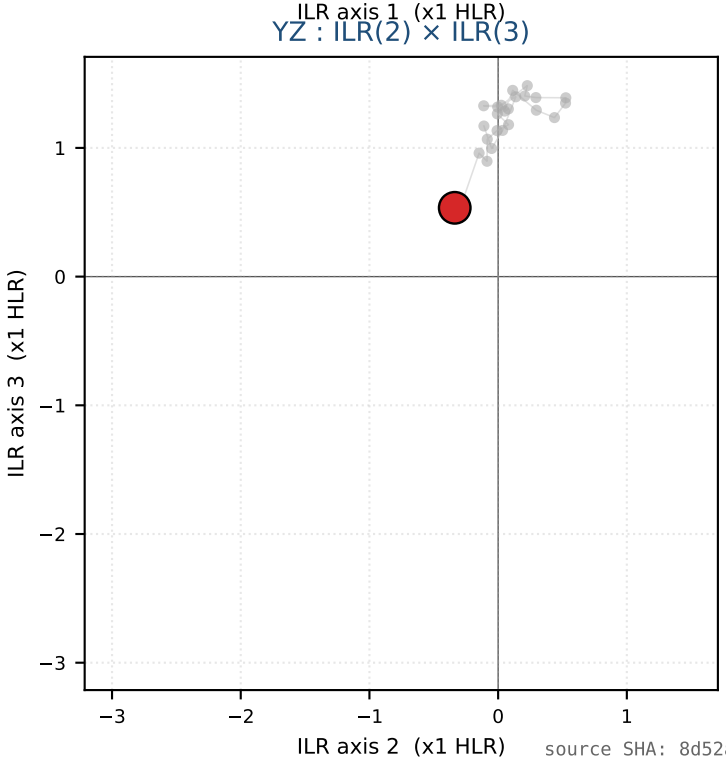
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

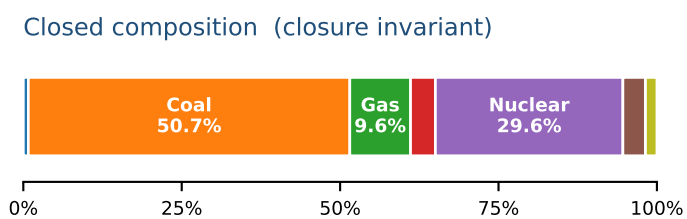




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0079	+2.855
Coal	0.5074	+7.014
Gas	0.0959	+5.348
Hydro	0.0393	+4.455
Nuclear	0.2959	+6.475
Other Fossil	0.0354	+4.353
Other Renewabl	0.0000	-33.207
Solar	0.0002	-0.971
Wind	0.0181	+3.679





closure: Σ = 1.000000 OK

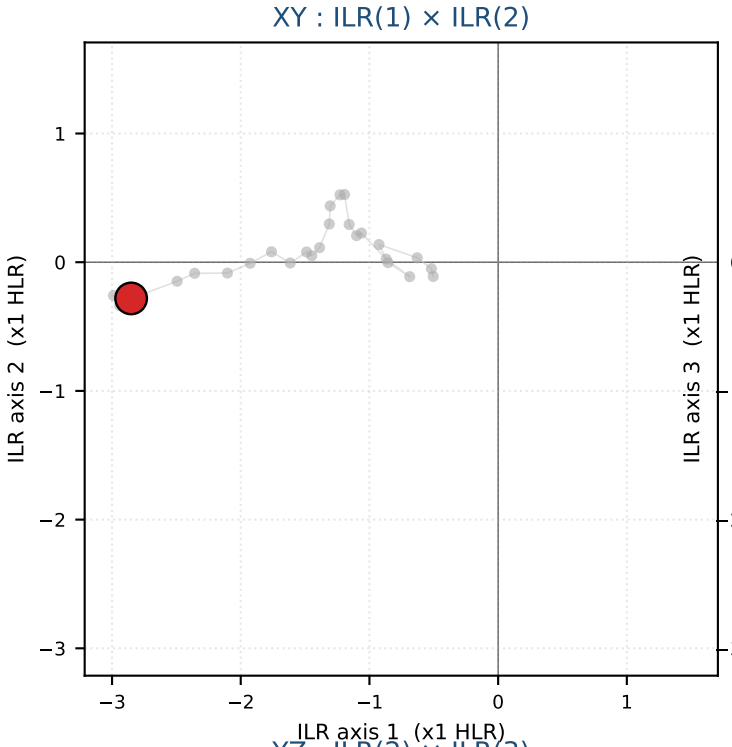
display factor: x1 (raw HLR x 1)

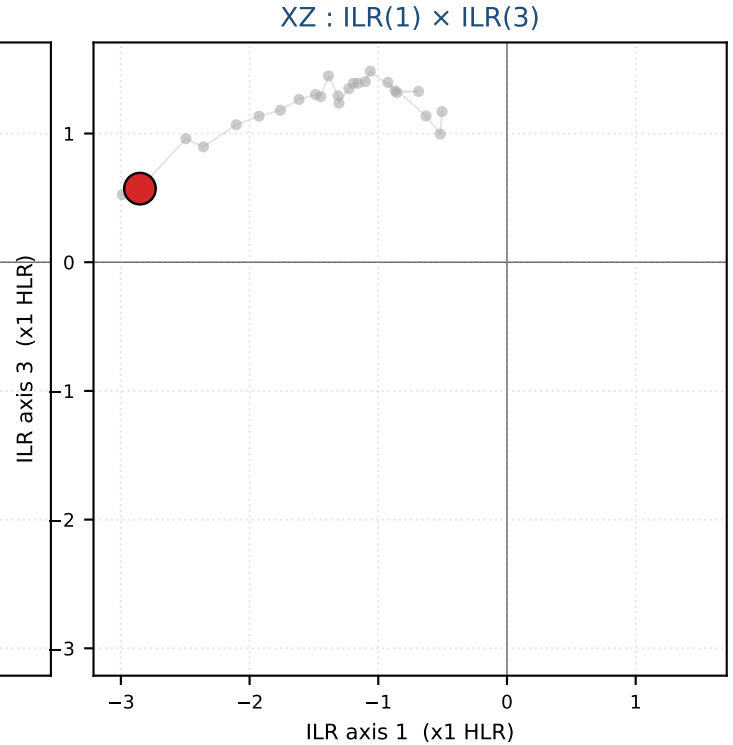
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=2 | 2002 | D=9 T=26 | level 1 first-principles

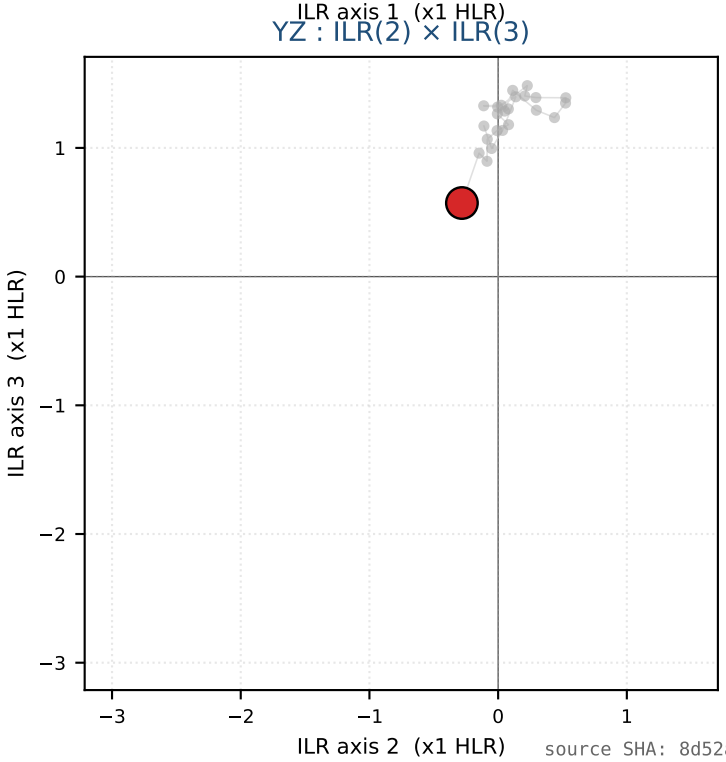
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

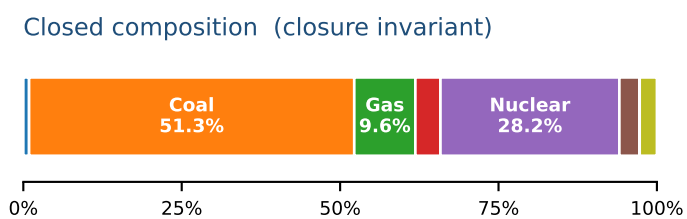




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0091	+2.871
Coal	0.5132	+6.904
Gas	0.0964	+5.233
Hydro	0.0396	+4.343
Nuclear	0.2824	+6.307
Other Fossil	0.0317	+4.121
Other Renewabl	0.0000	-33.337
Solar	0.0003	-0.408
Wind	0.0272	+3.966





closure: Σ = 1.000000 OK

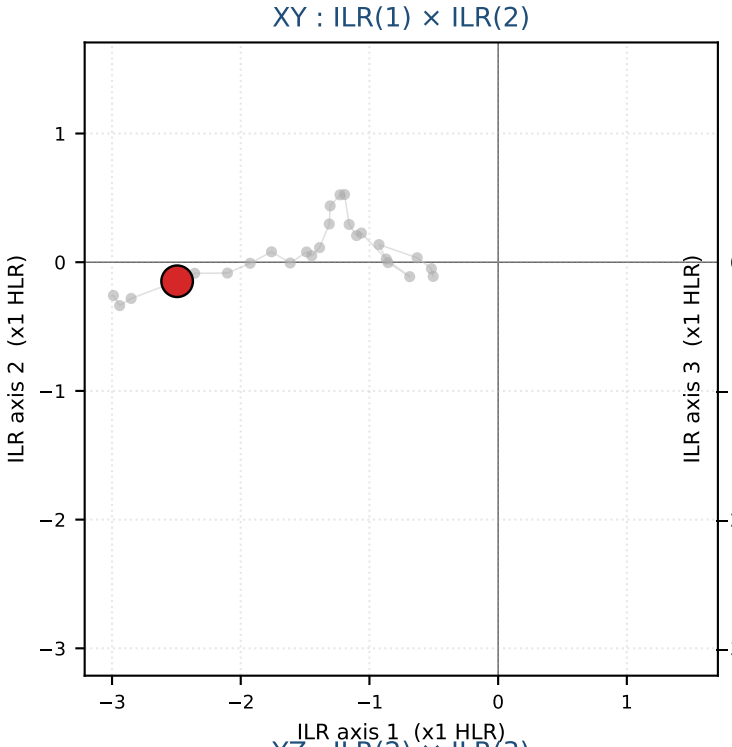
display factor: x1 (raw HLR x 1)

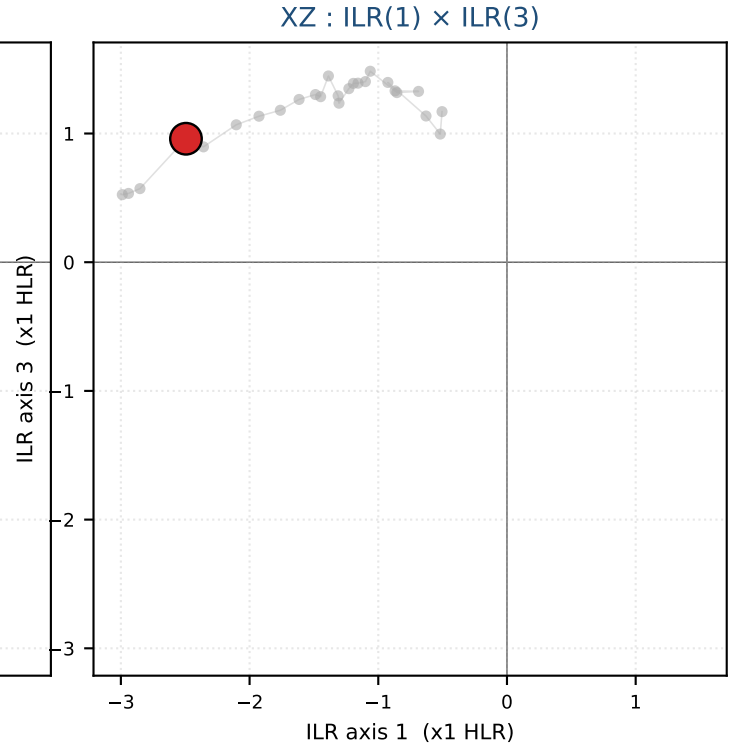
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=3 | 2003 | D=9 T=26 | level 1 first-principles

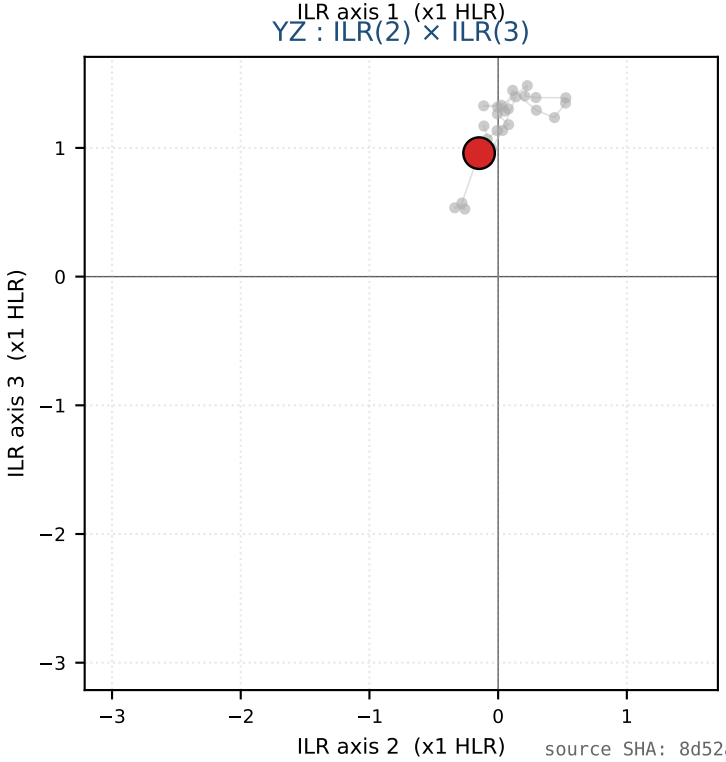
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



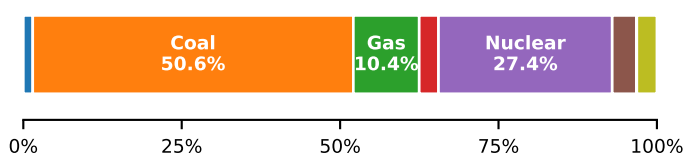


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0149	+3.254
Coal	0.5060	+6.781
Gas	0.1040	+5.199
Hydro	0.0304	+3.970
Nuclear	0.2742	+6.169
Other Fossil	0.0383	+4.201
Other Renewabl	0.0000	-33.477
Solar	0.0005	-0.109
Wind	0.0317	+4.011



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

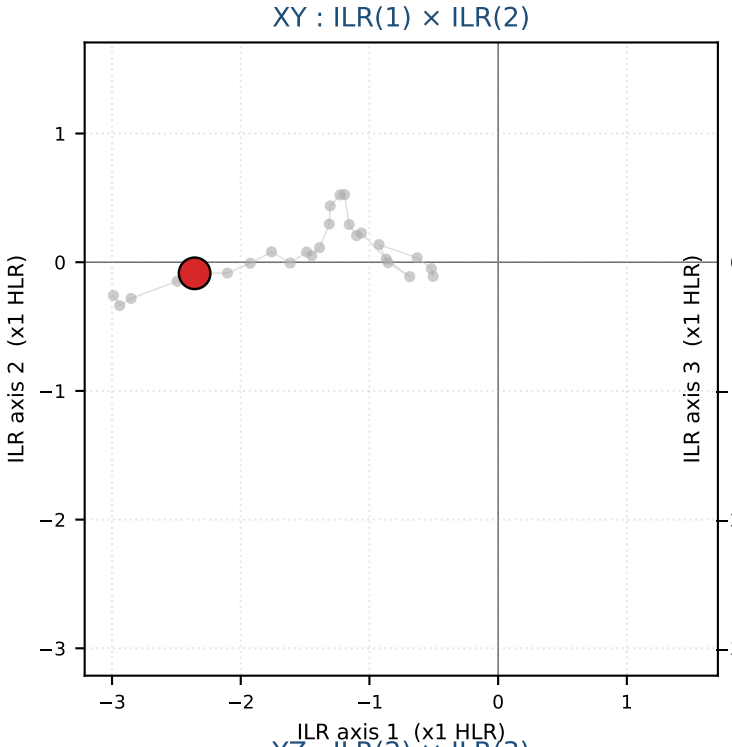
display factor: x1 (raw HLR x 1)

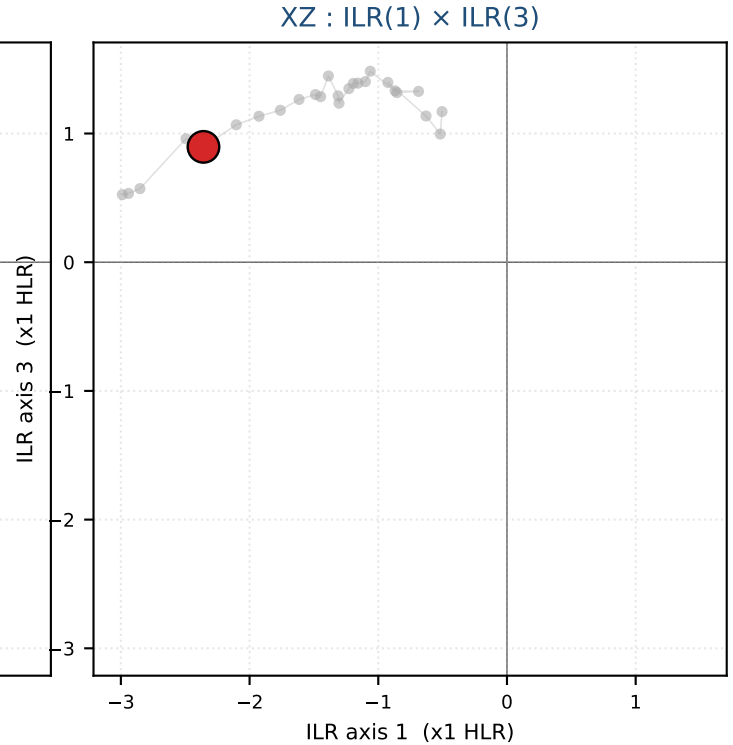
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=4 | 2004 | D=9 T=26 | level 1 first-principles

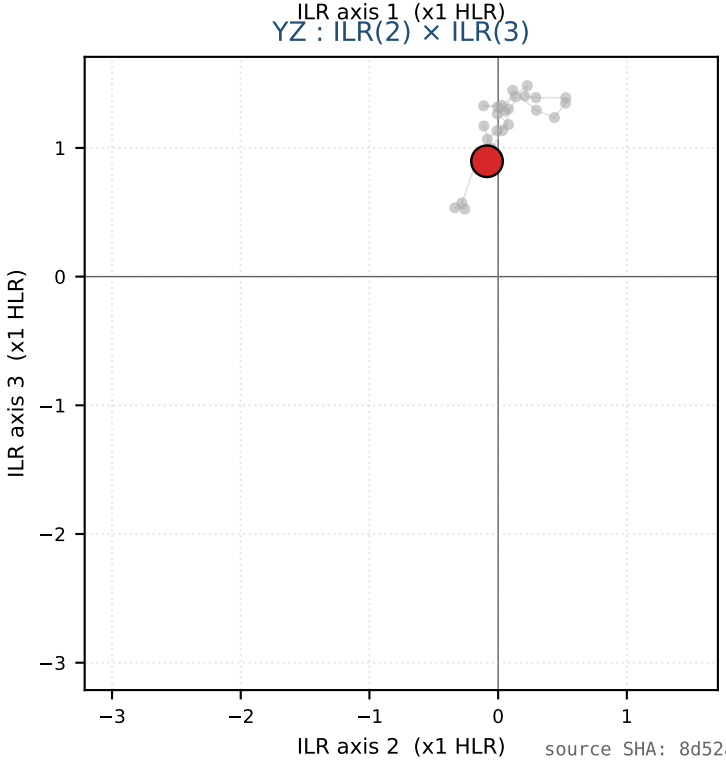
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



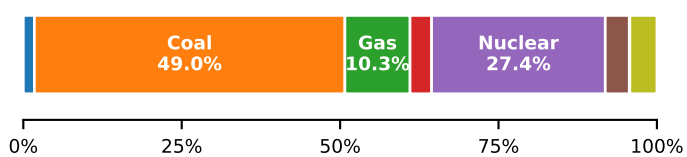


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0175	+3.294
Coal	0.4901	+6.629
Gas	0.1028	+5.067
Hydro	0.0340	+3.962
Nuclear	0.2741	+6.048
Other Fossil	0.0379	+4.070
Other Renewabl	0.0000	-33.609
Solar	0.0009	+0.350
Wind	0.0427	+4.188



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

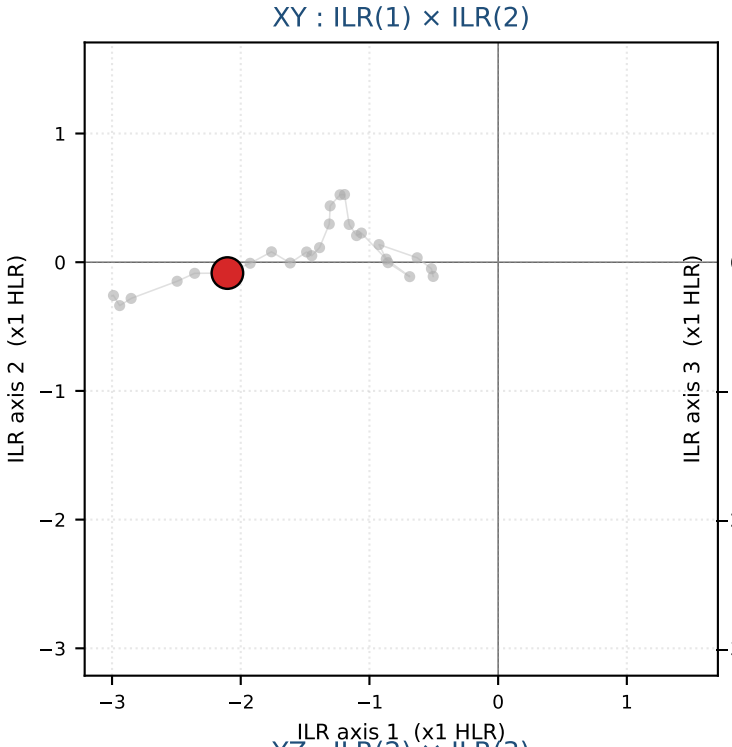
display factor: x1 (raw HLR x 1)

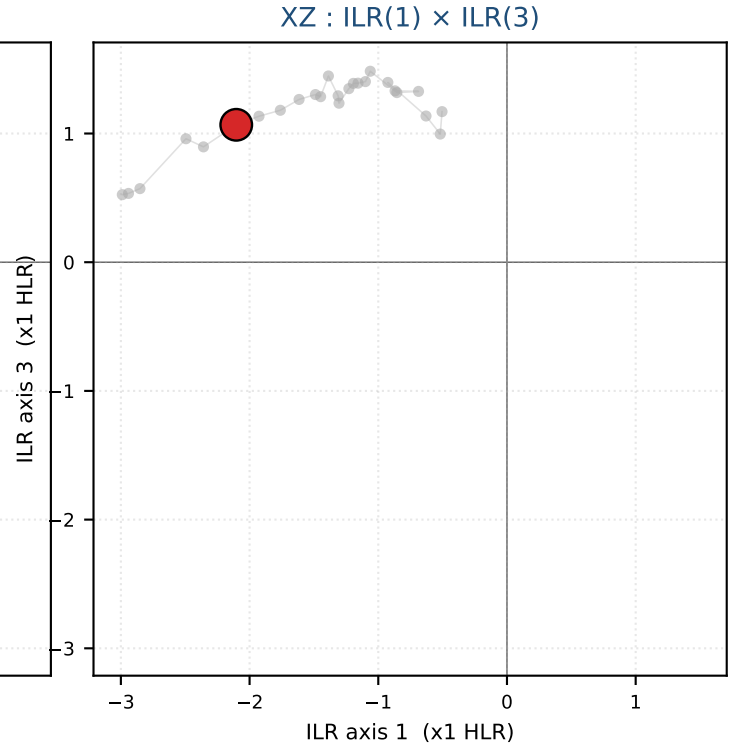
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=5 | 2005 | D=9 T=26 | level 1 first-principles

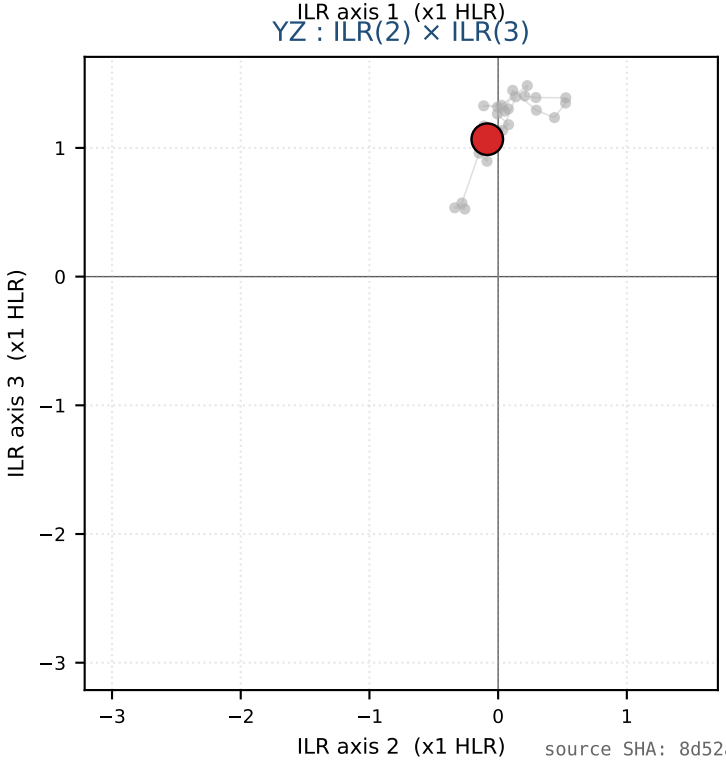
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



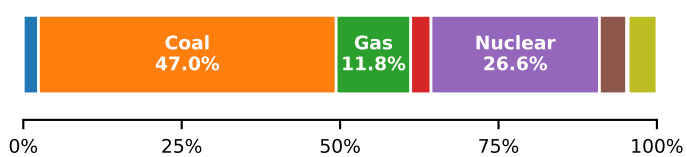


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0240	+3.462
Coal	0.4698	+6.437
Gas	0.1177	+5.053
Hydro	0.0320	+3.751
Nuclear	0.2658	+5.868
Other Fossil	0.0433	+4.052
Other Renewabl	0.0000	-33.765
Solar	0.0021	+1.044
Wind	0.0453	+4.098



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

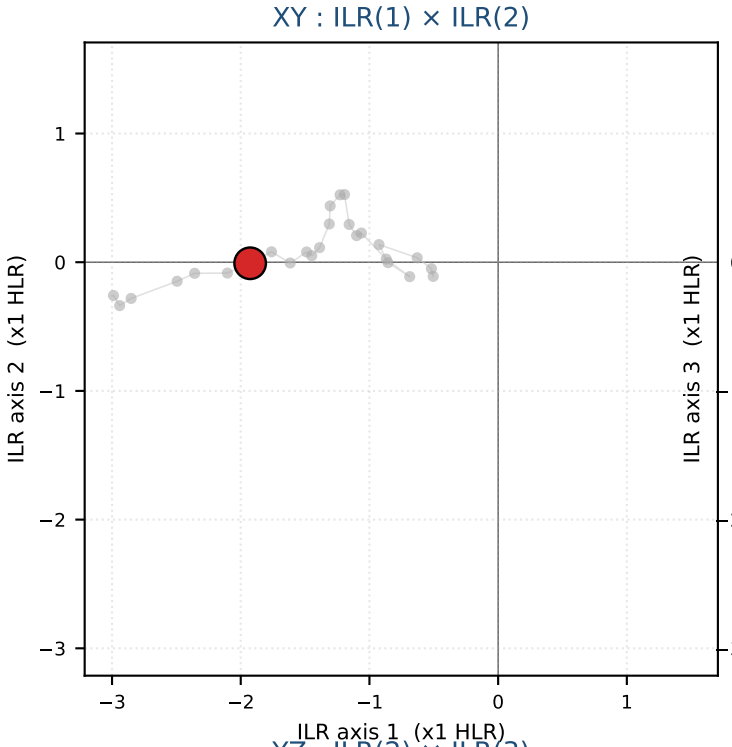
display factor: x1 (raw HLR x 1)

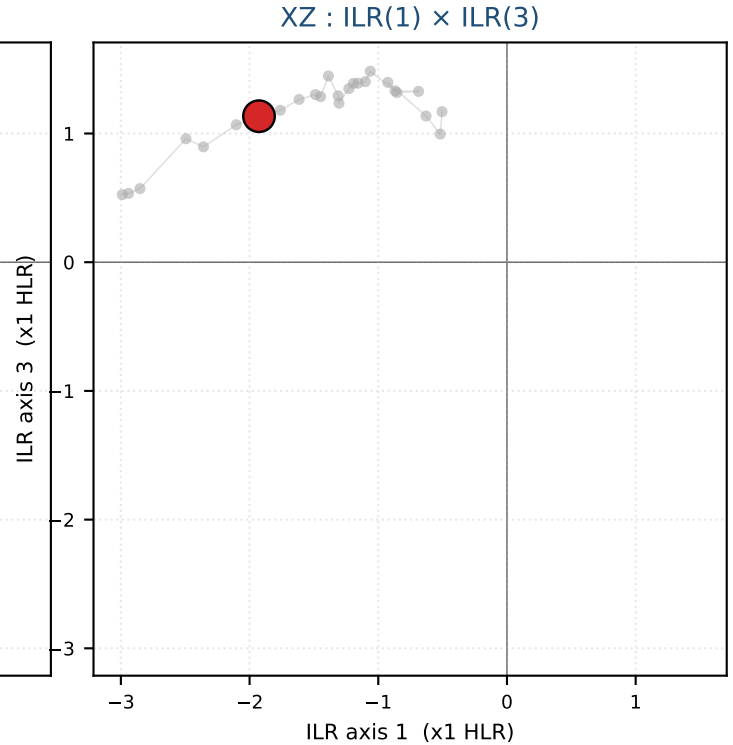
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=6 | 2006 | D=9 T=26 | level 1 first-principles

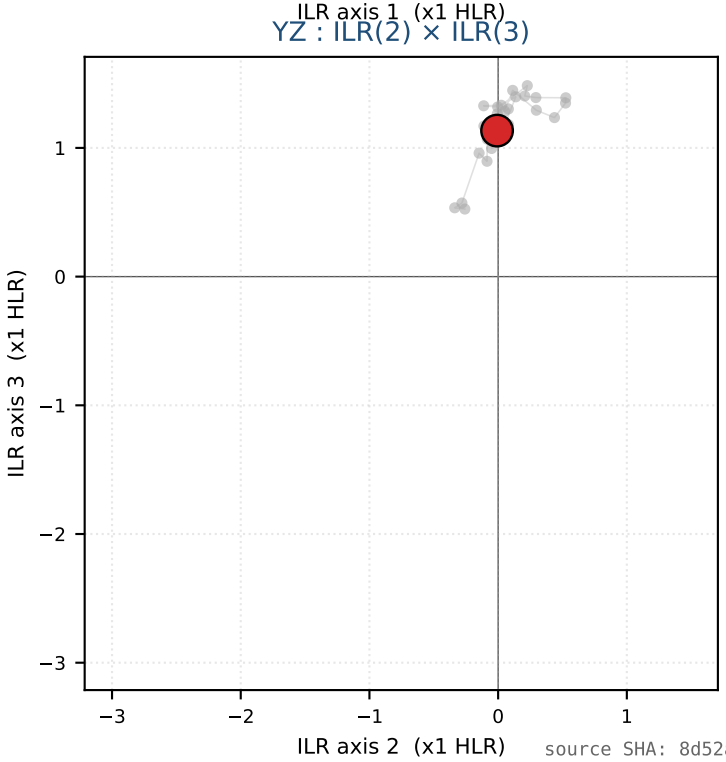
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



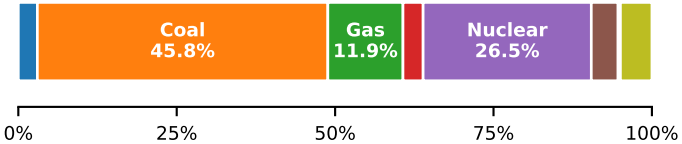


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0301	+3.603
Coal	0.4585	+6.327
Gas	0.1186	+4.975
Hydro	0.0318	+3.659
Nuclear	0.2654	+5.781
Other Fossil	0.0424	+3.946
Other Renewabl	0.0000	-33.877
Solar	0.0036	+1.481
Wind	0.0497	+4.106



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

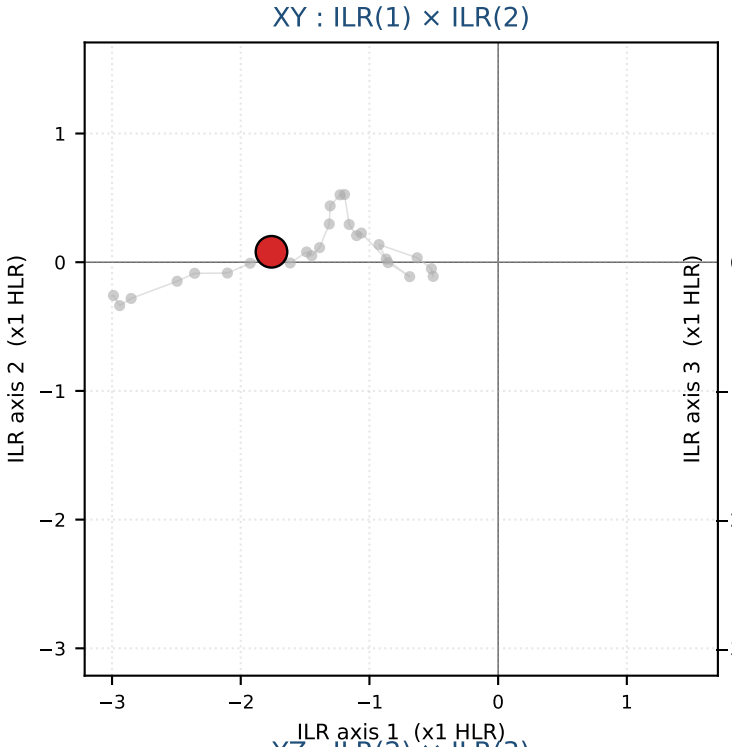
display factor: x1 (raw HLR x 1)

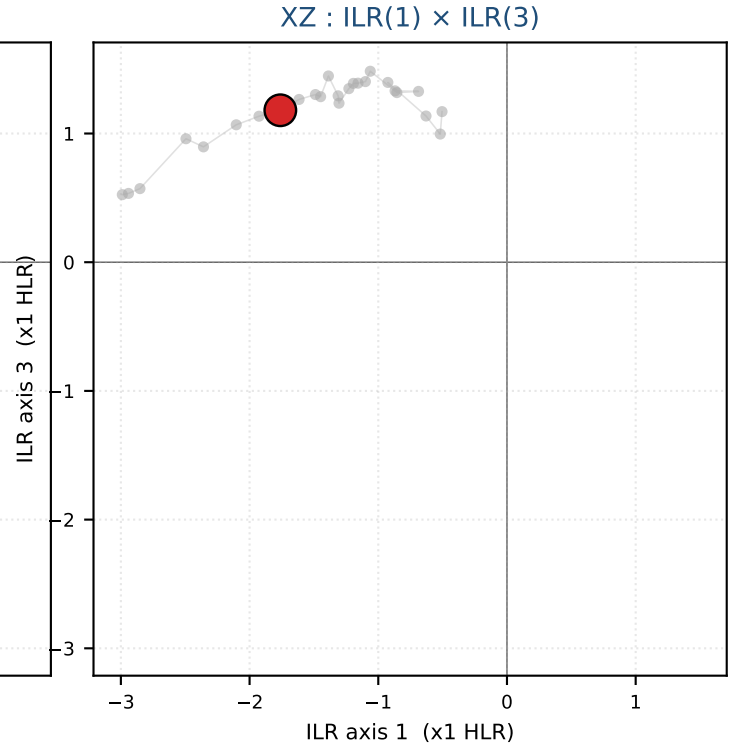
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=7 | 2007 | D=9 T=26 | level 1 first-principles

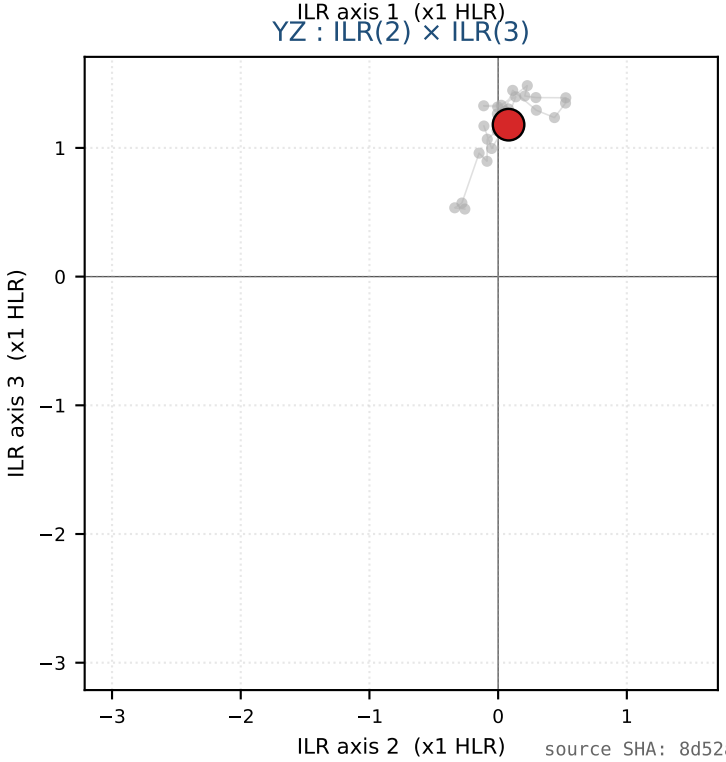
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



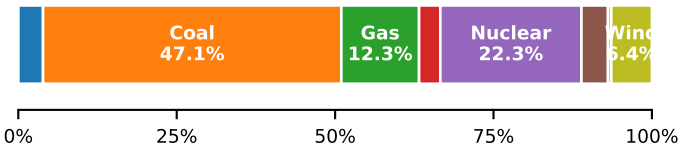


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0390	+3.778
Coal	0.4708	+6.268
Gas	0.1227	+4.924
Hydro	0.0335	+3.627
Nuclear	0.2227	+5.520
Other Fossil	0.0421	+3.854
Other Renewabl	0.0000	-33.965
Solar	0.0050	+1.718
Wind	0.0642	+4.276



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

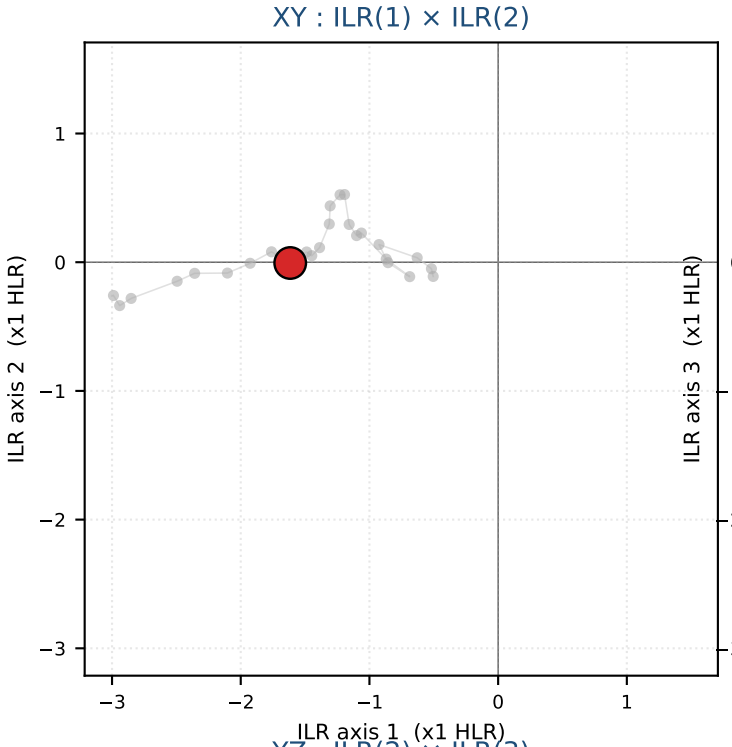
display factor: x1 (raw HLR x 1)

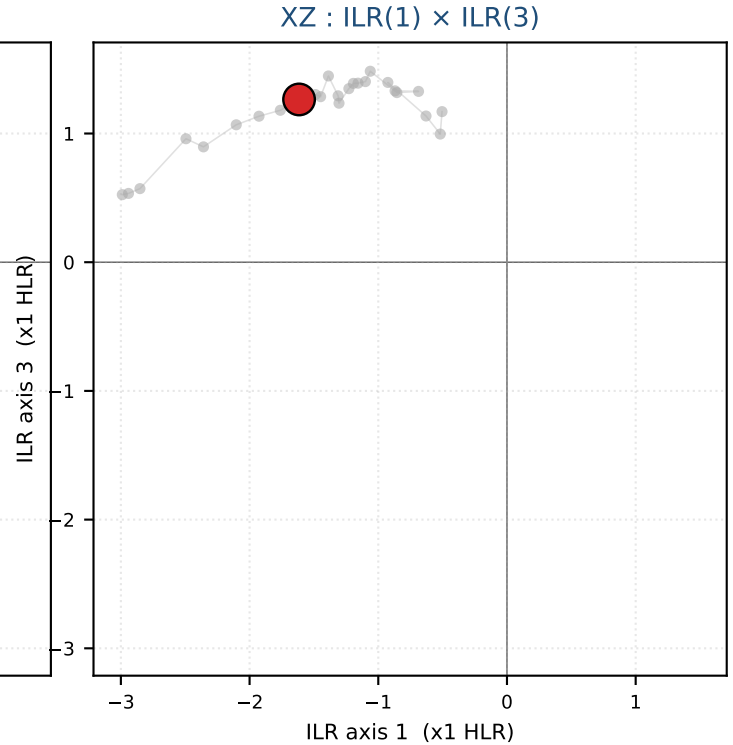
Helmert basis loadings (top 3 carriers per axis)

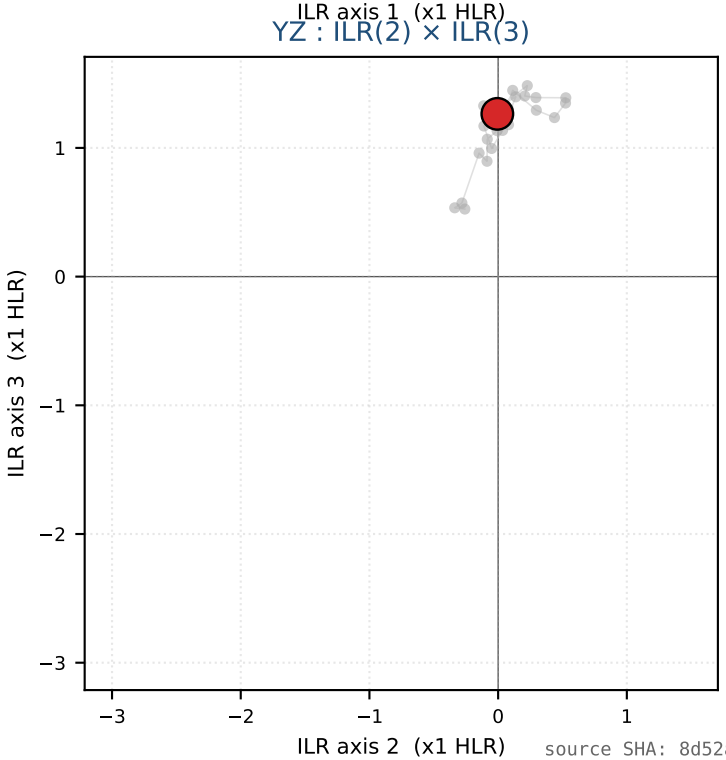
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=8 | 2008 | D=9 T=26 | level 1 first-principles

ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



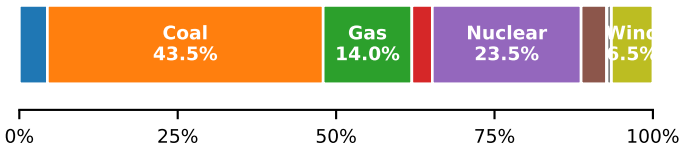




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0443	+0.442
Coal	0.4353	+2.727
Gas	0.1399	+1.592
Hydro	0.0323	+0.127
Nuclear	0.2348	+2.110
Other Fossil	0.0407	+0.357
Other Renewabl	0.0000	-6.803
Solar	0.0071	-1.384
Wind	0.0654	+0.832

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

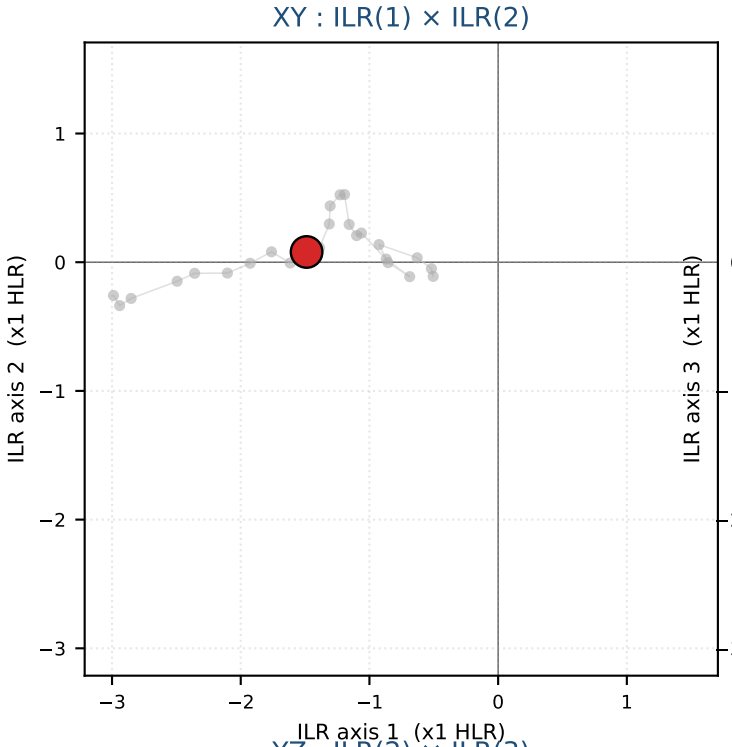
display factor: x1 (raw HLR x 1)

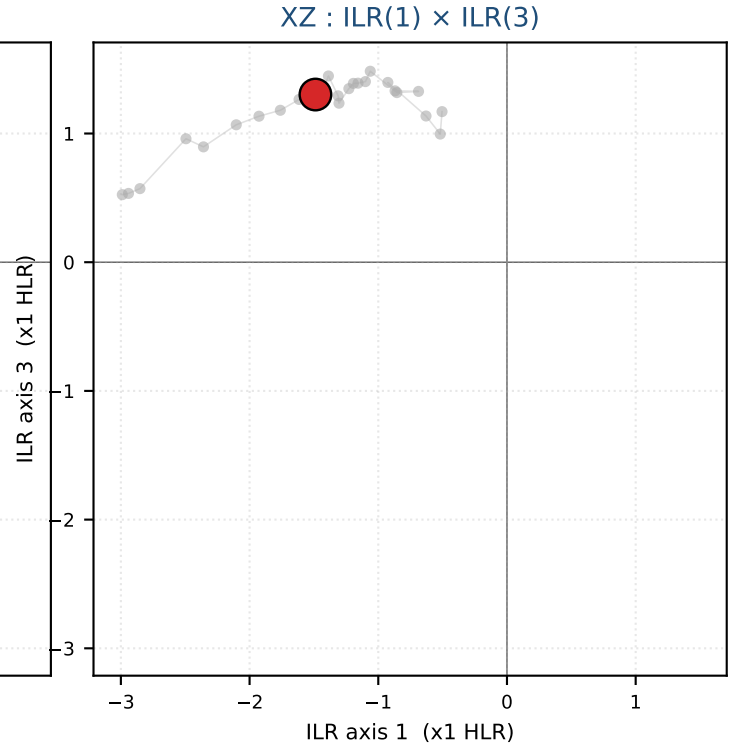
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=9 | 2009 | D=9 T=26 | level 1 first-principles

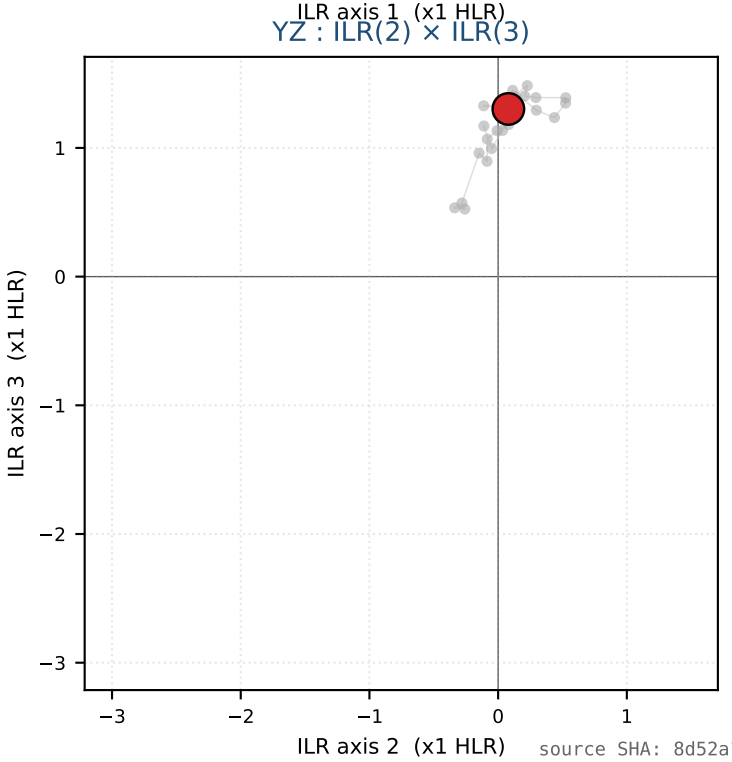
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

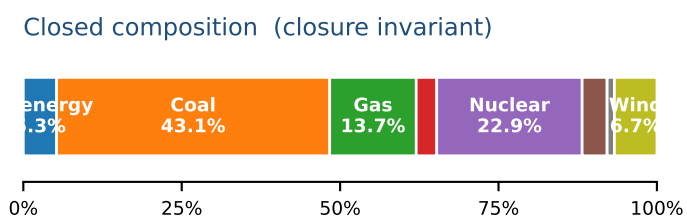




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0525	+0.540
Coal	0.4311	+2.645
Gas	0.1365	+1.495
Hydro	0.0324	+0.056
Nuclear	0.2295	+2.014
Other Fossil	0.0395	+0.255
Other Renewabl	0.0000	-6.803
Solar	0.0114	-0.985
Wind	0.0670	+0.784





closure: Σ = 1.000000 OK

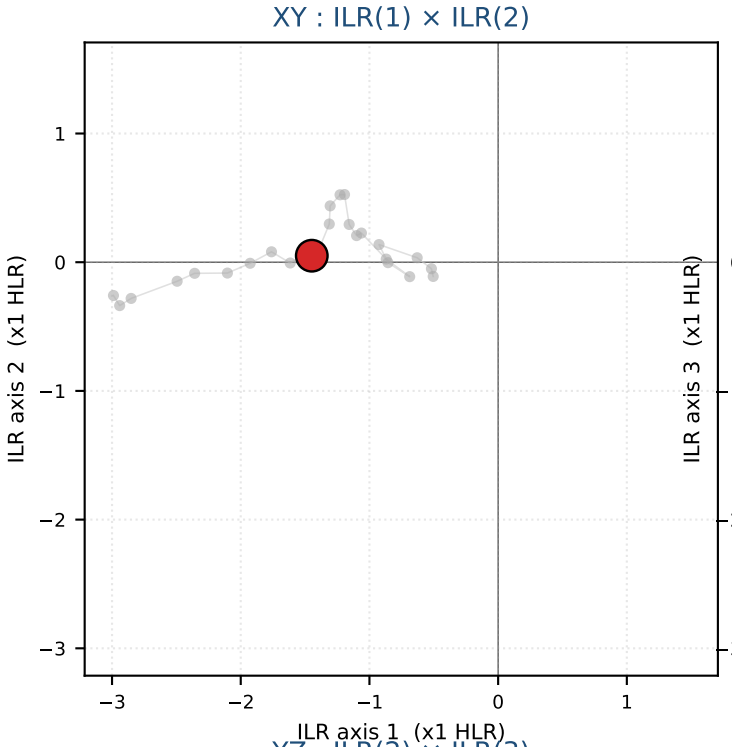
display factor: x1 (raw HLR x 1)

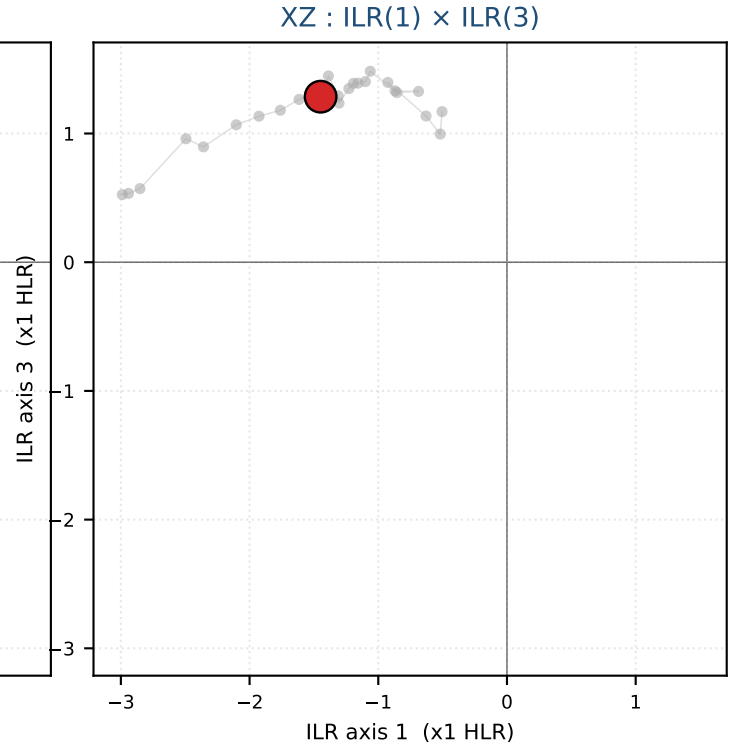
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=10 | 2010 | D=9 T=26 | level 1 first-principles

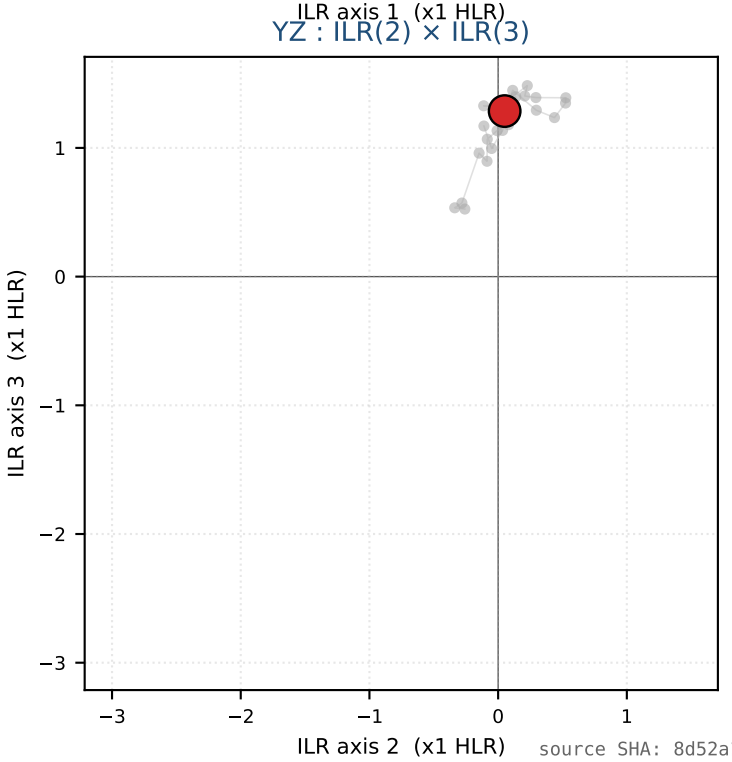
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

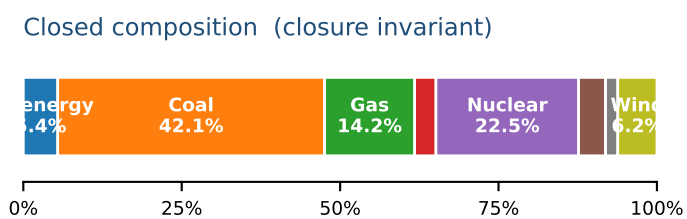




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0543	+0.471
Coal	0.4212	+2.519
Gas	0.1422	+1.433
Hydro	0.0336	-0.011
Nuclear	0.2252	+1.893
Other Fossil	0.0426	+0.228
Other Renewabl	0.0000	-6.560
Solar	0.0192	-0.571
Wind	0.0618	+0.599





closure: Σ = 1.000000 OK

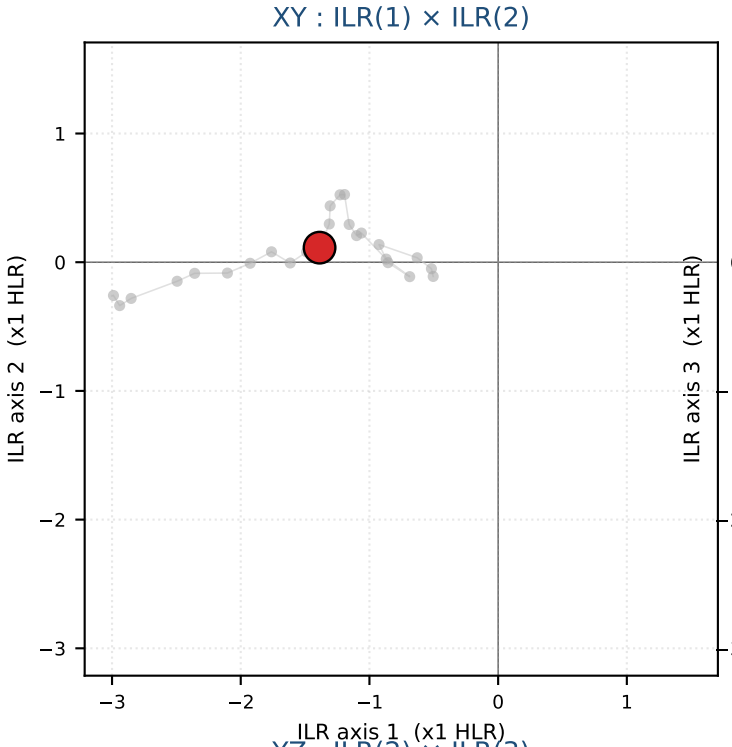
display factor: x1 (raw HLR x 1)

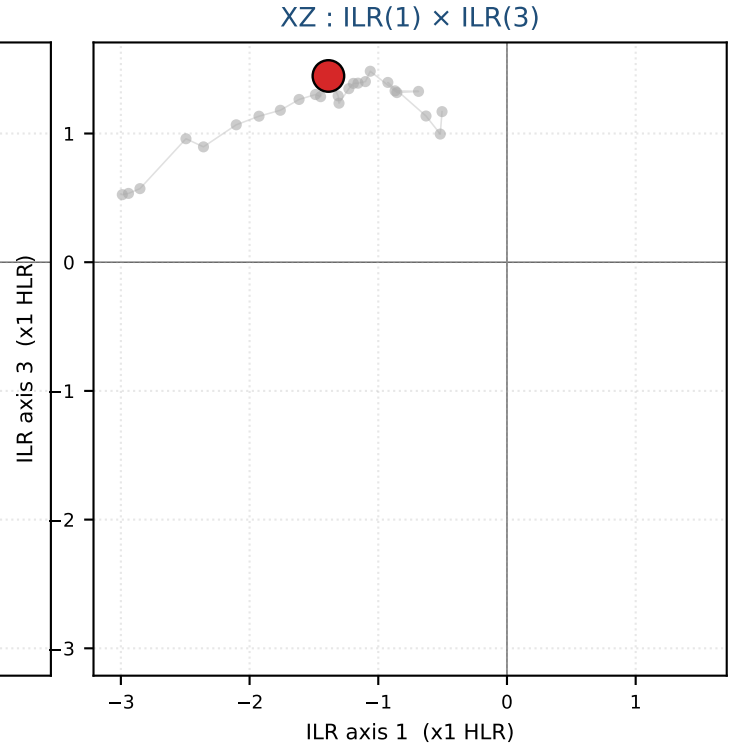
Helmert basis loadings (top 3 carriers per axis)

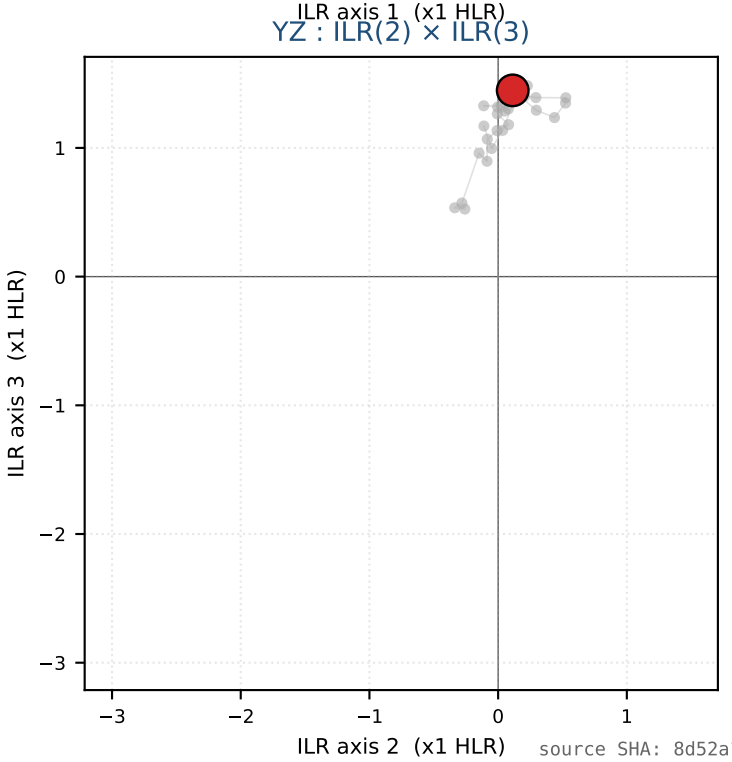
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=11 | 2011 | D=9 T=26 | level 1 first-principles

ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



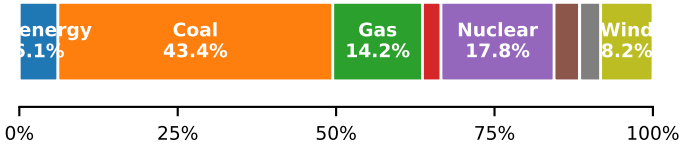




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0610	+0.567
Coal	0.4338	+2.529
Gas	0.1416	+1.409
Hydro	0.0292	-0.169
Nuclear	0.1785	+1.641
Other Fossil	0.0404	+0.156
Other Renewabl	0.0000	-6.953
Solar	0.0330	-0.046
Wind	0.0824	+0.868

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

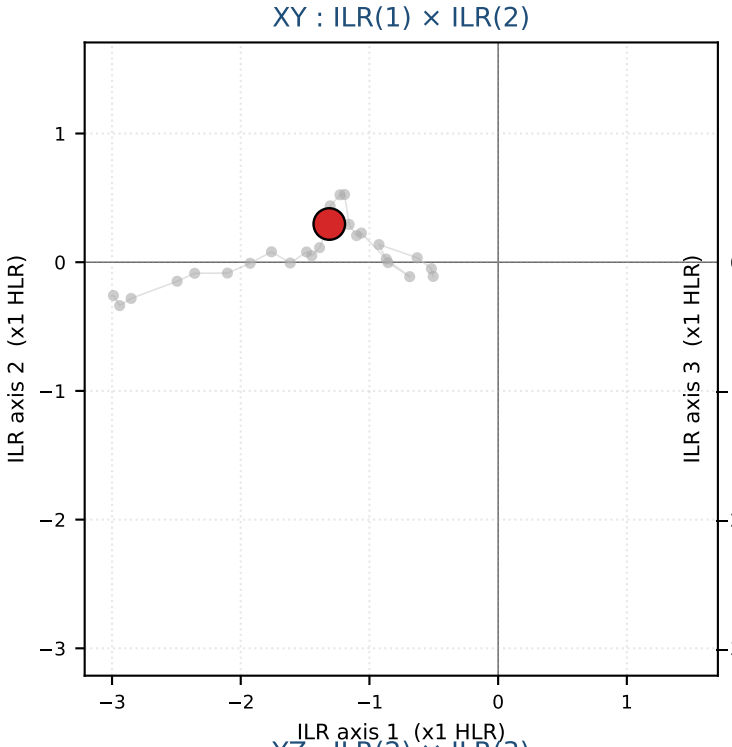
display factor: x1 (raw HLR x 1)

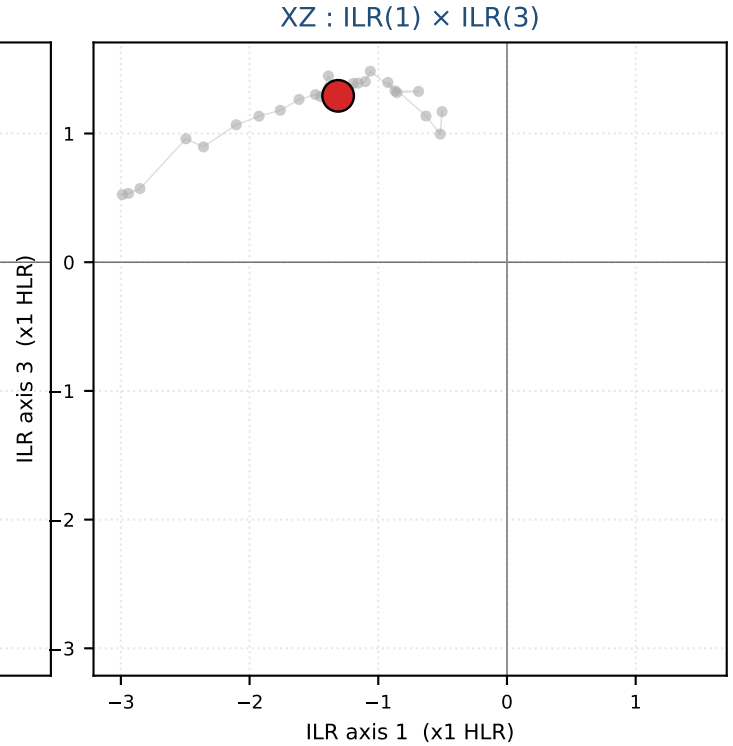
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=12 | 2012 | D=9 T=26 | level 1 first-principles

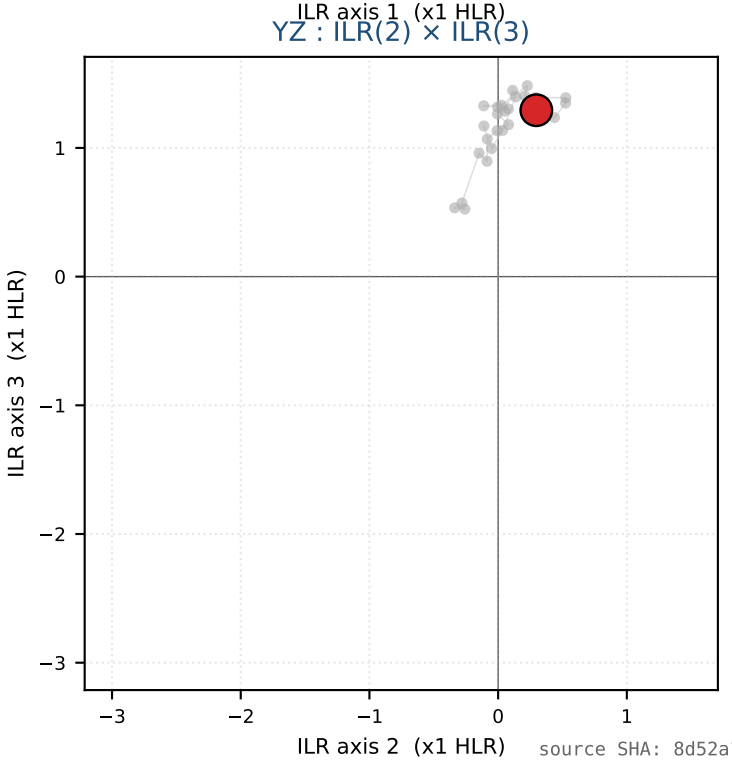
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



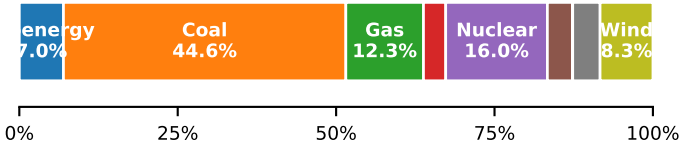


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0697	+0.618
Coal	0.4457	+2.473
Gas	0.1225	+1.182
Hydro	0.0351	-0.068
Nuclear	0.1605	+1.452
Other Fossil	0.0400	+0.062
Other Renewabl	0.0000	-6.654
Solar	0.0431	+0.138
Wind	0.0834	+0.797



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

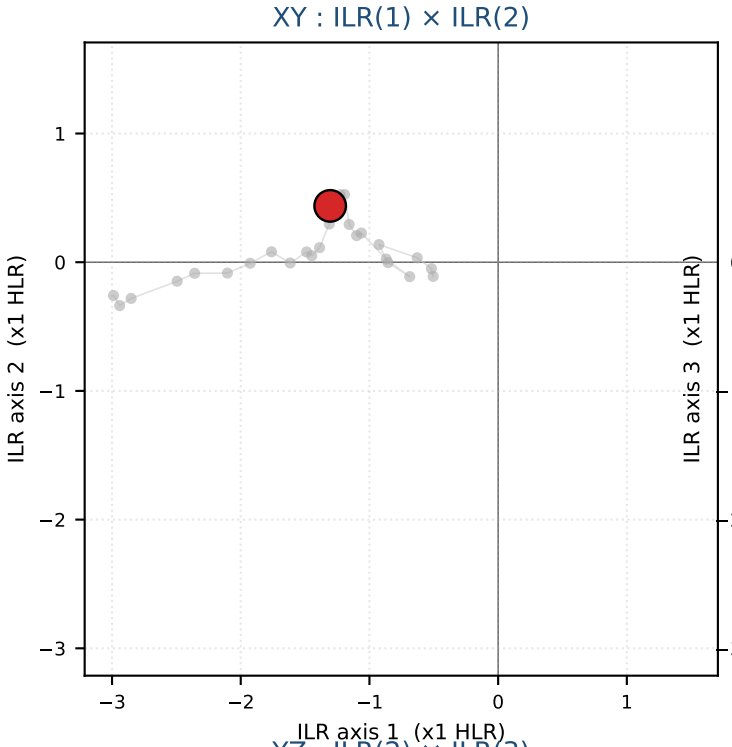
display factor: x1 (raw HLR x 1)

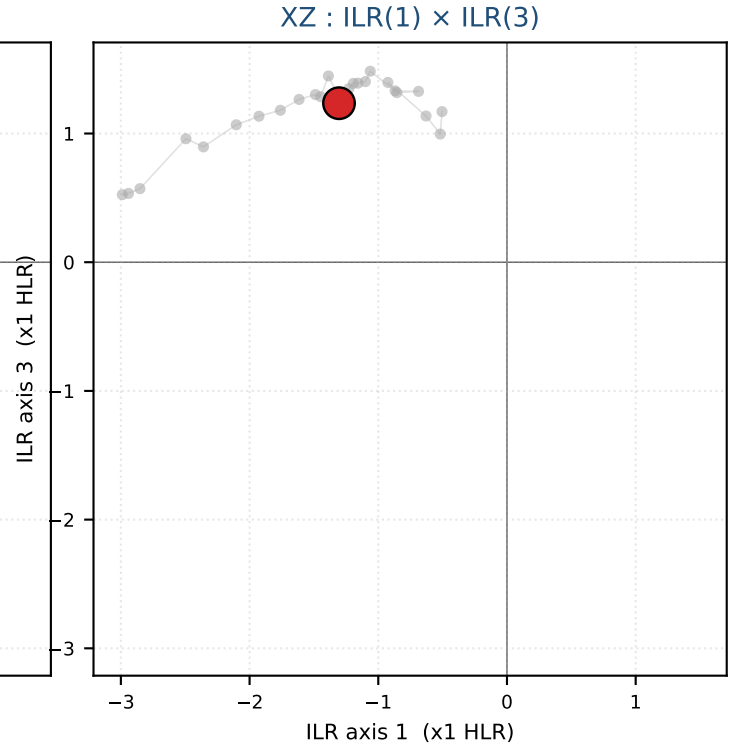
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=13 | 2013 | D=9 T=26 | level 1 first-principles

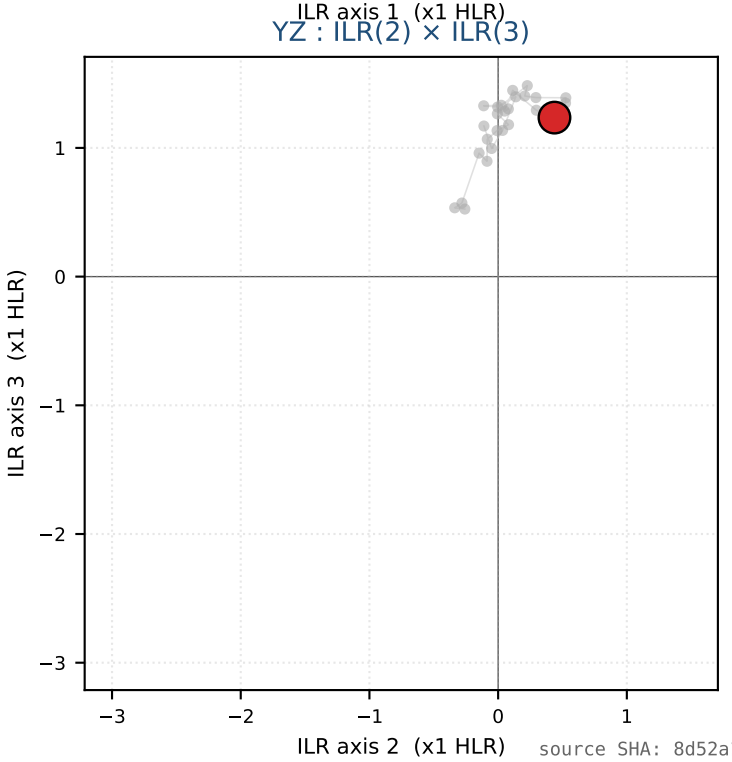
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



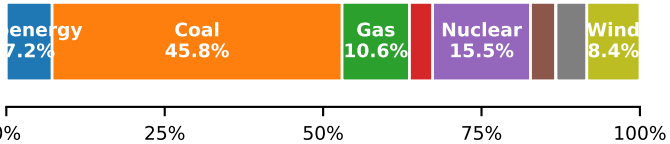


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0723	+0.541
Coal	0.4577	+2.387
Gas	0.1064	+0.928
Hydro	0.0365	-0.141
Nuclear	0.1545	+1.301
Other Fossil	0.0401	-0.048
Other Renewabl	0.0001	-5.802
Solar	0.0486	+0.145
Wind	0.0838	+0.689



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

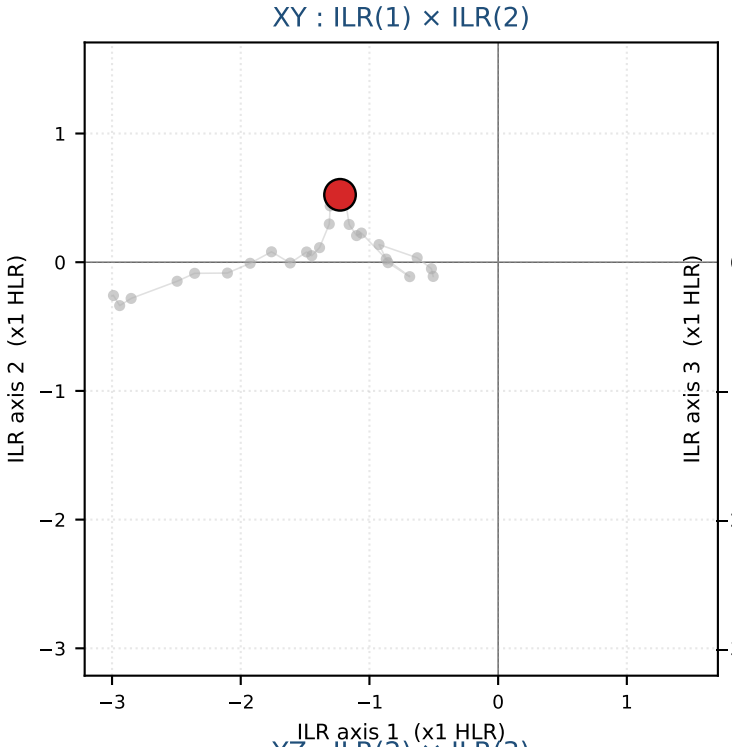
display factor: x1 (raw HLR x 1)

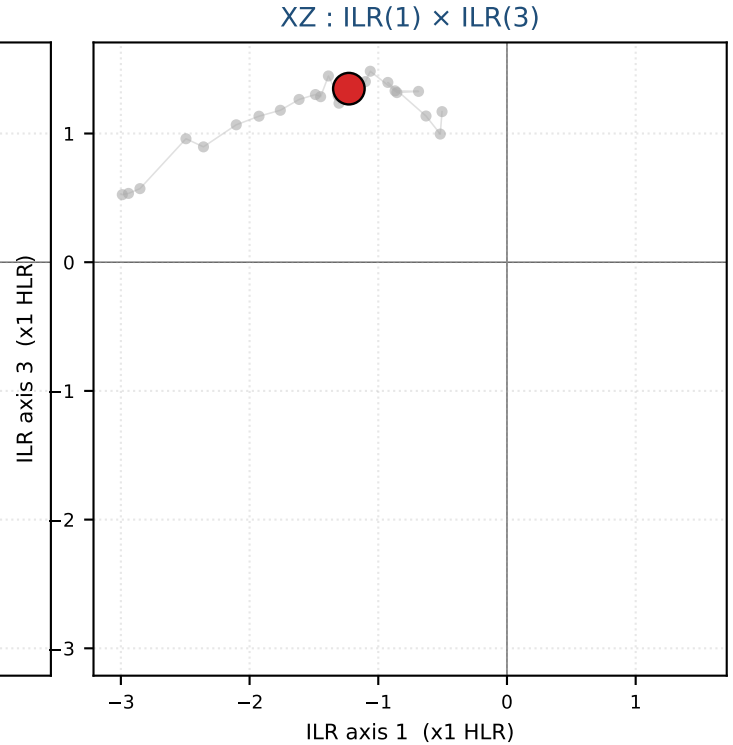
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=14 | 2014 | D=9 T=26 | level 1 first-principles

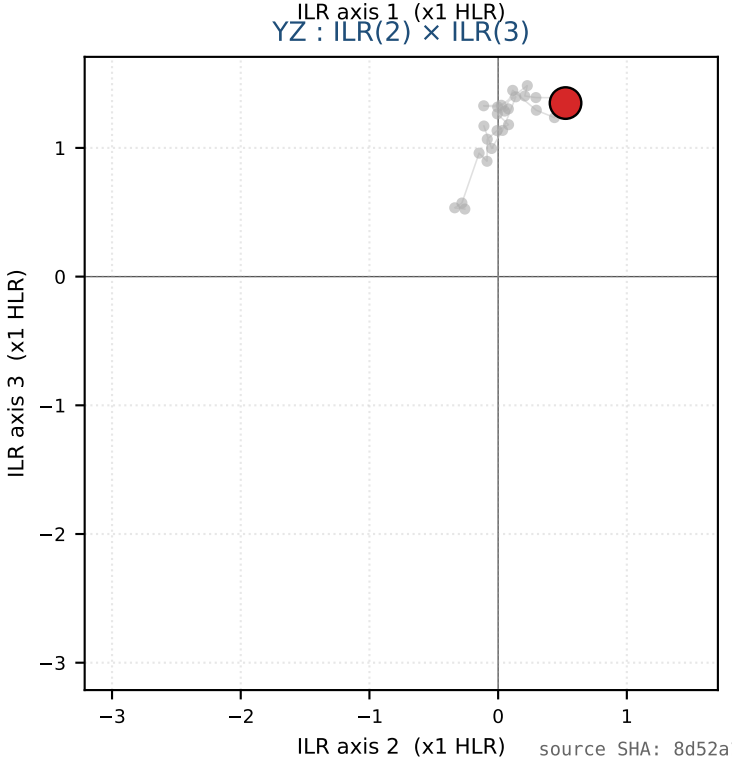
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

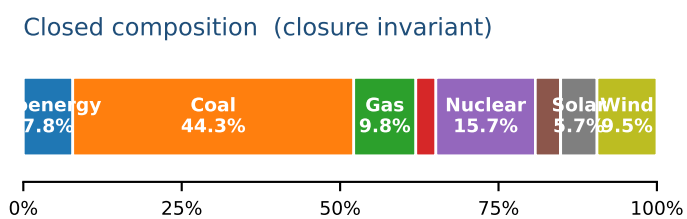




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0780	+0.579
Coal	0.4435	+2.316
Gas	0.0980	+0.806
Hydro	0.0317	-0.324
Nuclear	0.1570	+1.277
Other Fossil	0.0399	-0.093
Other Renewabl	0.0002	-5.601
Solar	0.0573	+0.270
Wind	0.0945	+0.770





closure: Σ = 1.000000 OK

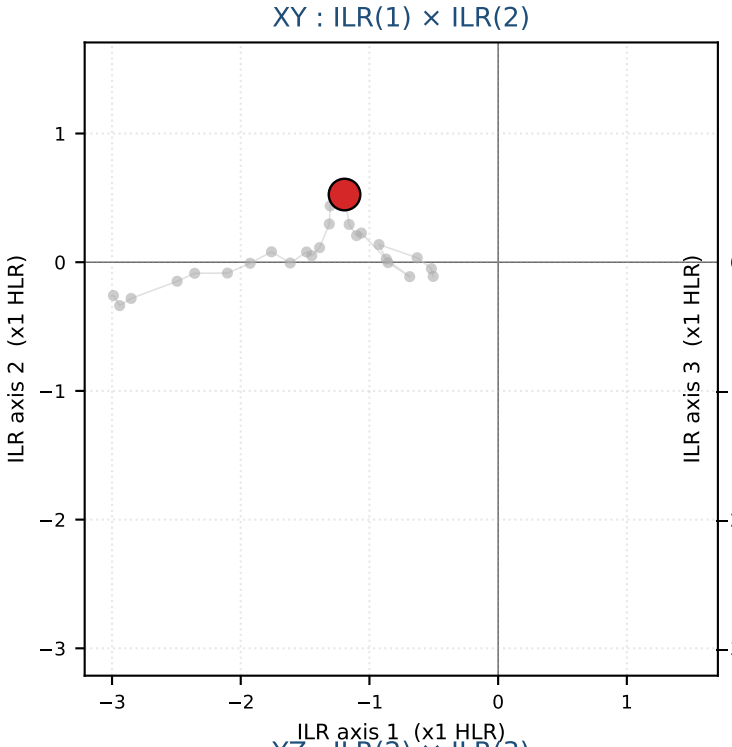
display factor: x1 (raw HLR x 1)

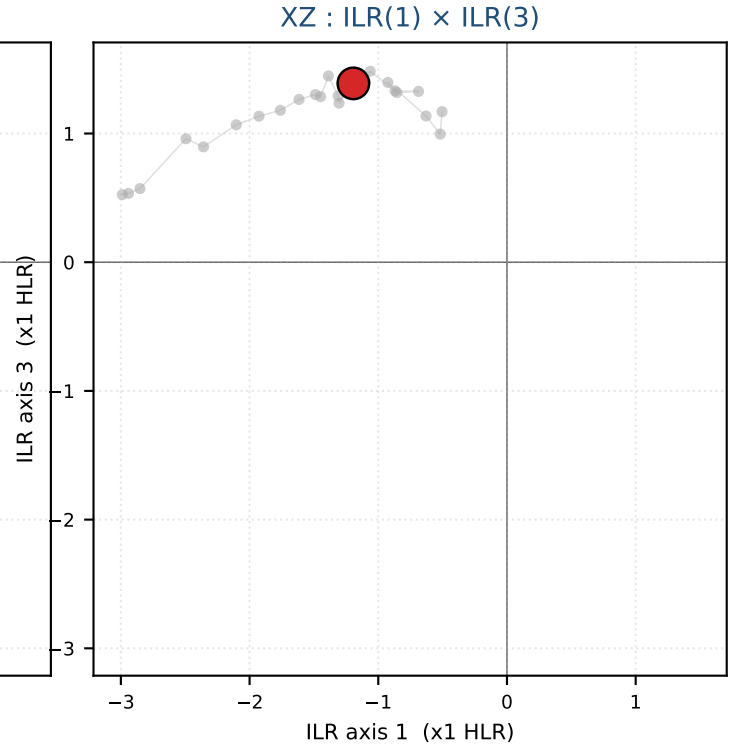
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=15 | 2015 | D=9 T=26 | level 1 first-principles

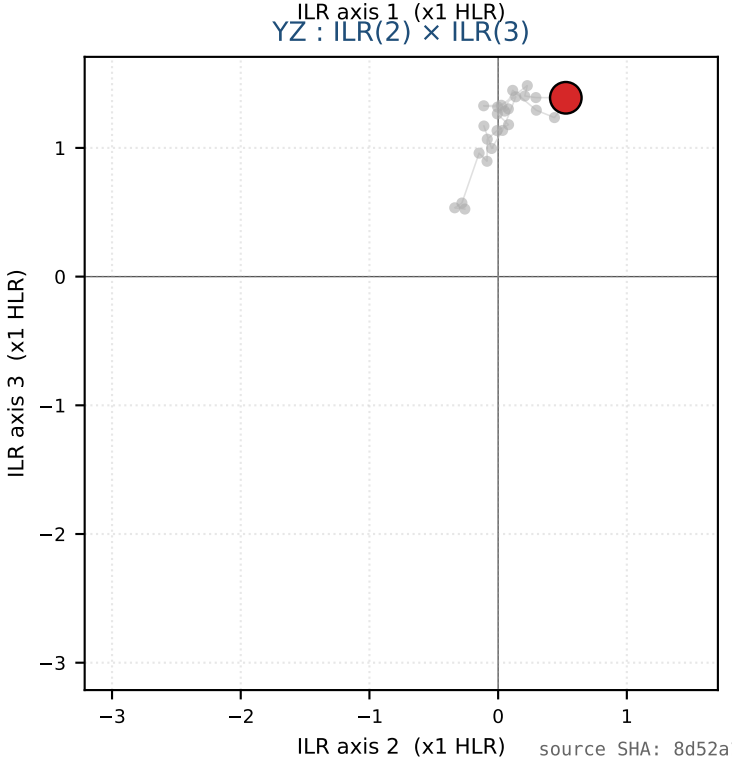
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



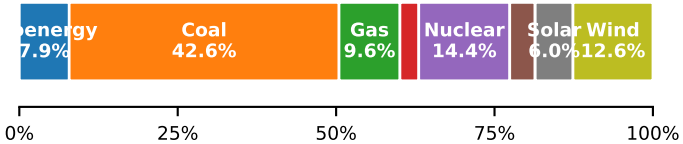


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0787	+0.548
Coal	0.4259	+2.236
Gas	0.0962	+0.748
Hydro	0.0297	-0.427
Nuclear	0.1436	+1.149
Other Fossil	0.0400	-0.129
Other Renewabl	0.0002	-5.411
Solar	0.0596	+0.269
Wind	0.1261	+1.019



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

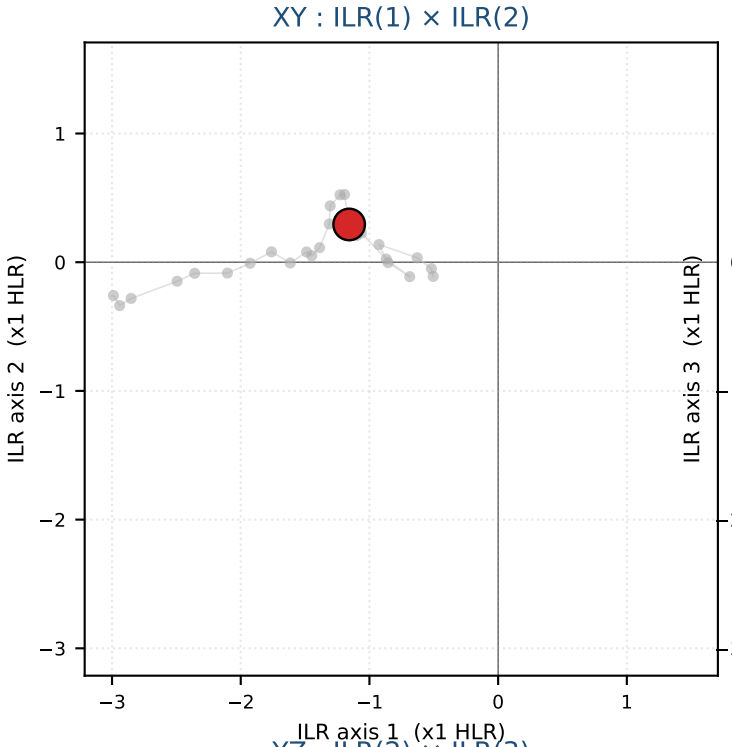
display factor: x1 (raw HLR x 1)

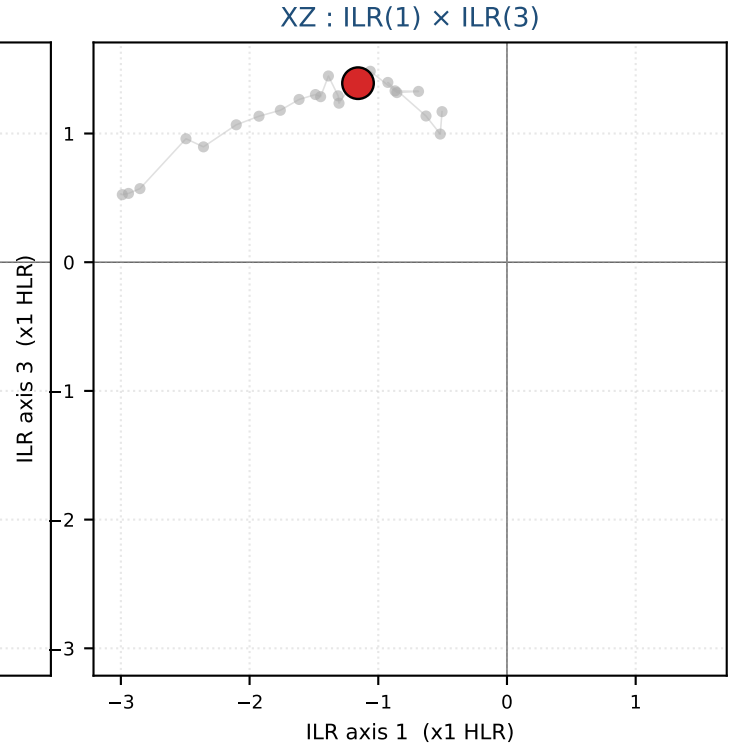
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=16 | 2016 | D=9 T=26 | level 1 first-principles

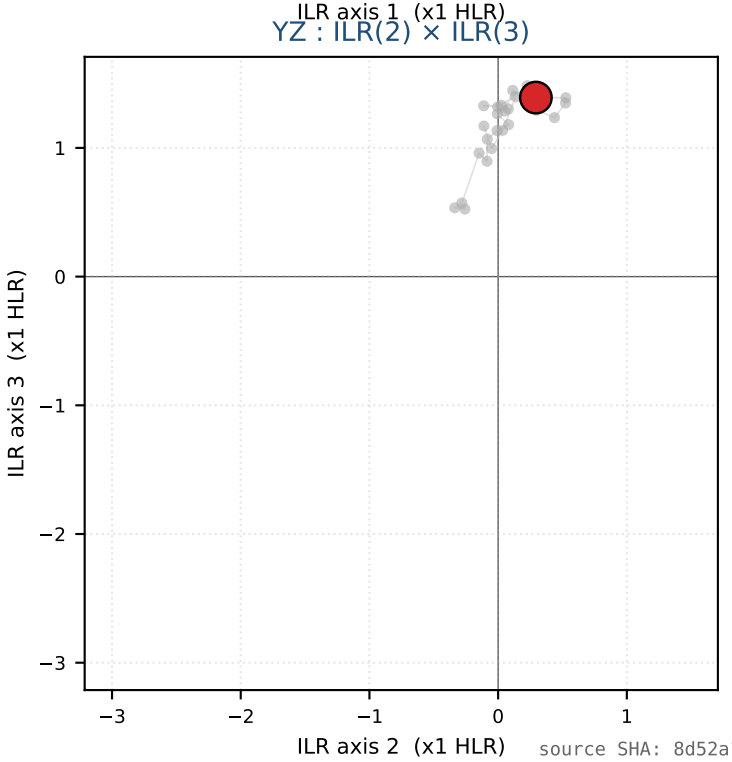
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

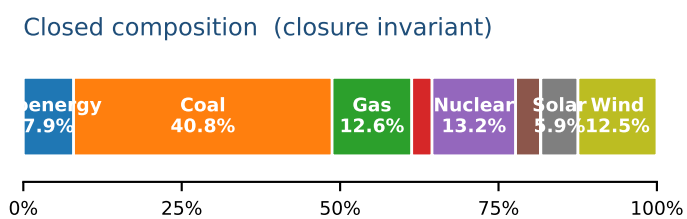




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0794	+0.506
Coal	0.4080	+2.143
Gas	0.1257	+0.966
Hydro	0.0320	-0.401
Nuclear	0.1319	+1.014
Other Fossil	0.0396	-0.190
Other Renewabl	0.0003	-5.196
Solar	0.0585	+0.202
Wind	0.1246	+0.957





closure: Σ = 1.000000 OK

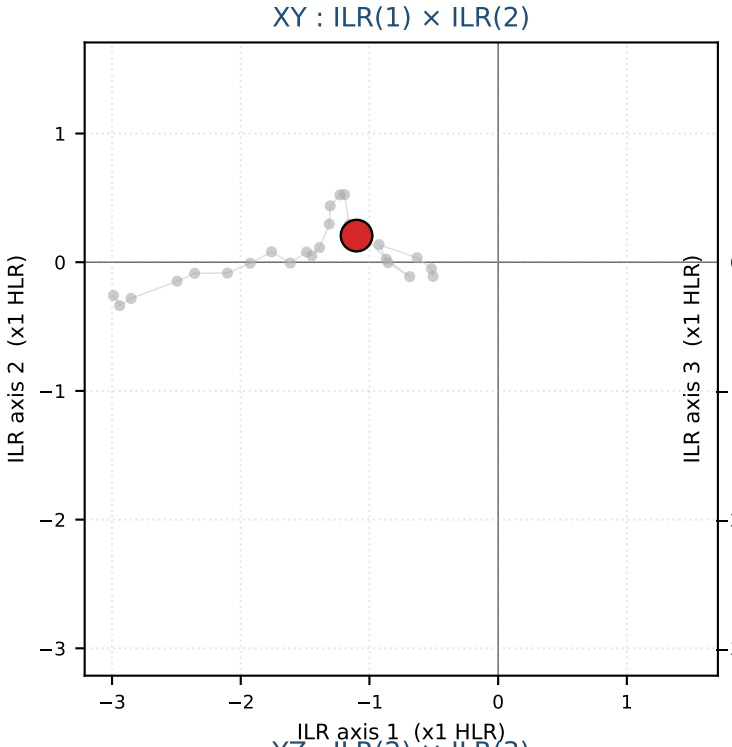
display factor: x1 (raw HLR x 1)

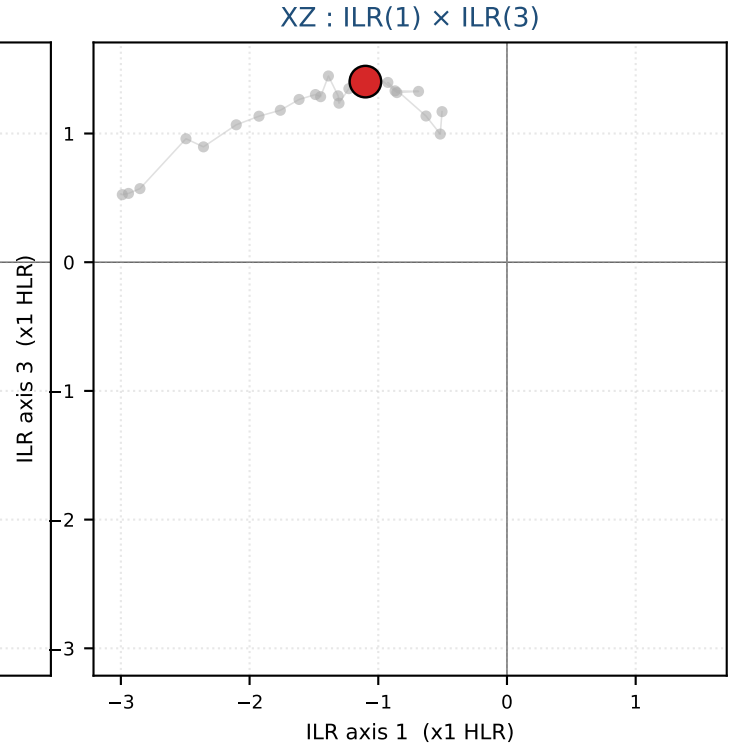
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=17 | 2017 | D=9 T=26 | level 1 first-principles

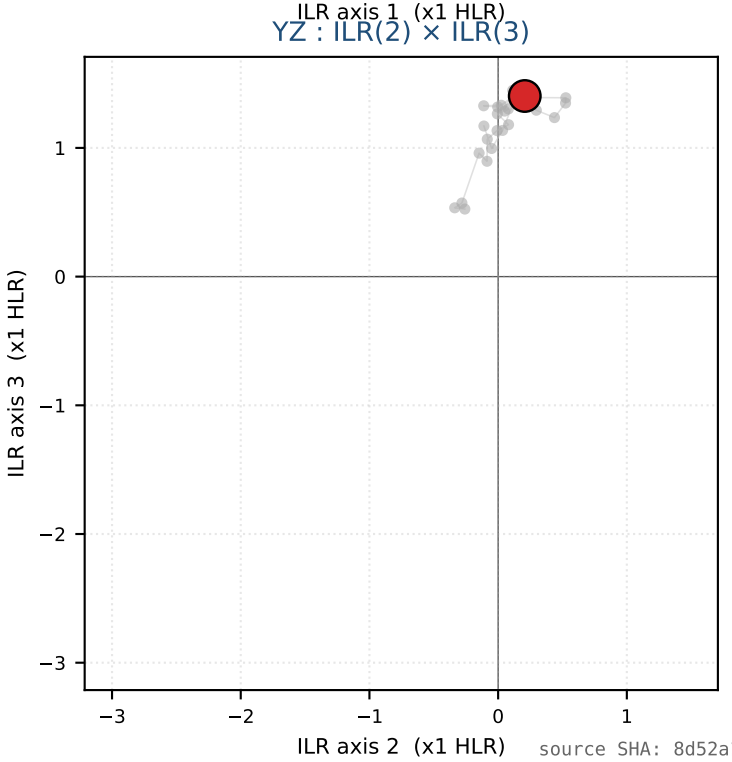
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

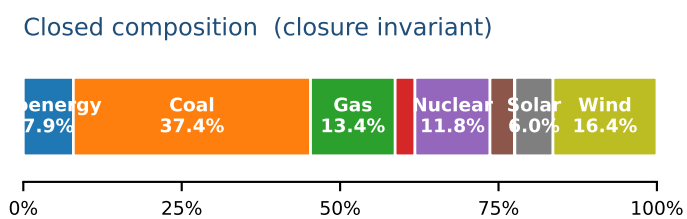




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0790	+0.495
Coal	0.3744	+2.050
Gas	0.1335	+1.019
Hydro	0.0313	-0.432
Nuclear	0.1184	+0.899
Other Fossil	0.0390	-0.210
Other Renewabl	0.0002	-5.268
Solar	0.0601	+0.222
Wind	0.1640	+1.225





closure: Σ = 1.000000 OK

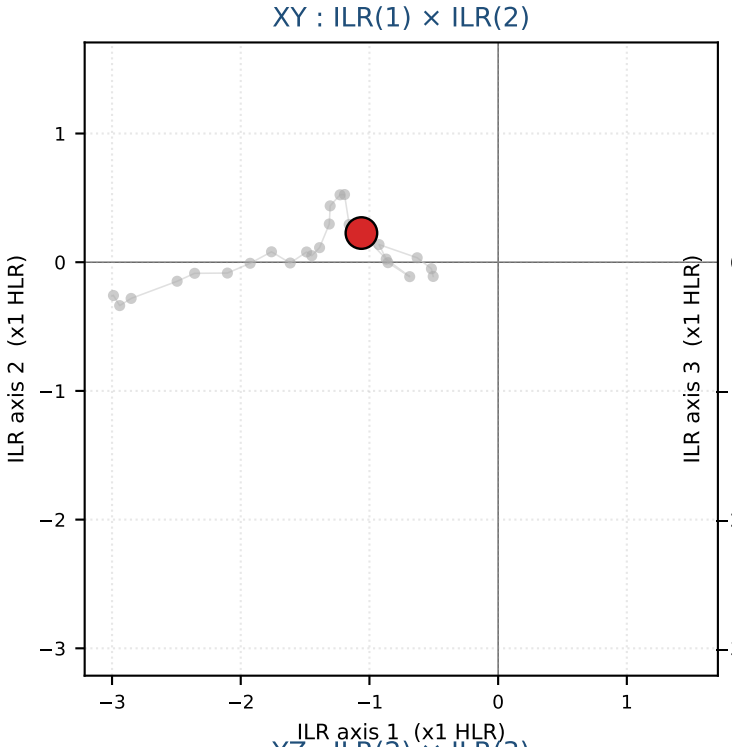
display factor: x1 (raw HLR x 1)

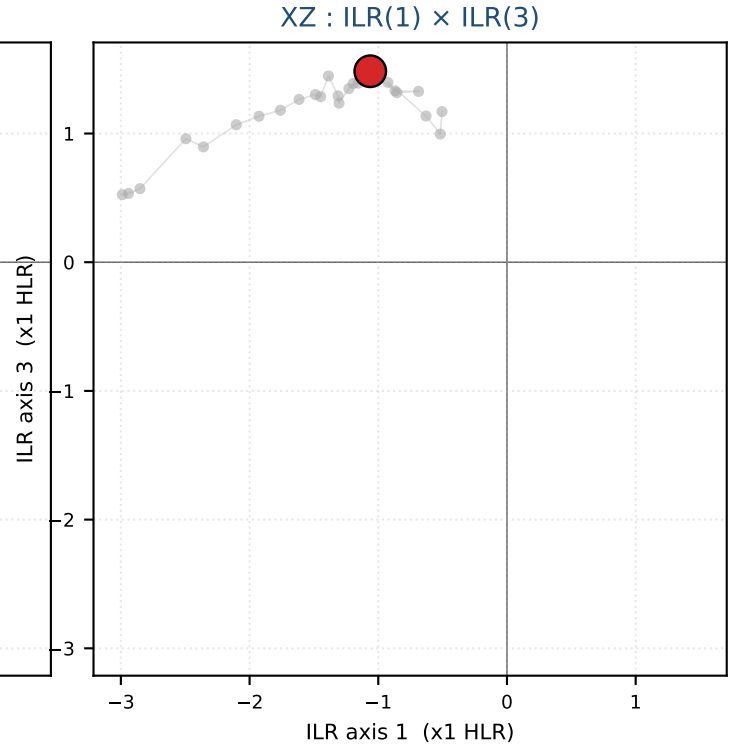
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=18 | 2018 | D=9 T=26 | level 1 first-principles

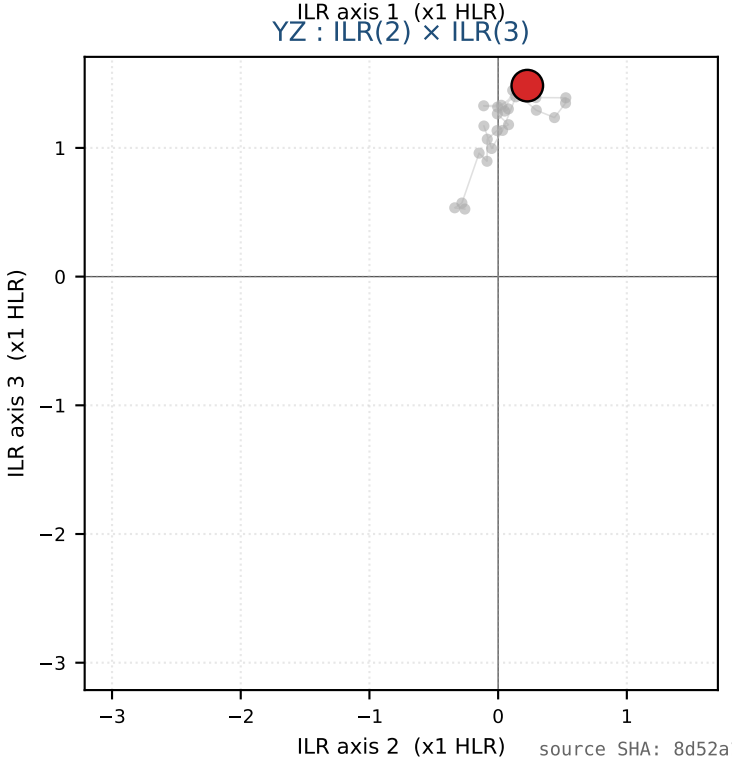
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



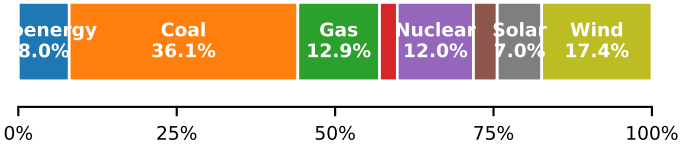


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0803	+0.494
Coal	0.3609	+1.996
Gas	0.1290	+0.968
Hydro	0.0280	-0.561
Nuclear	0.1202	+0.897
Other Fossil	0.0374	-0.271
Other Renewabl	0.0003	-5.148
Solar	0.0701	+0.358
Wind	0.1739	+1.266



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

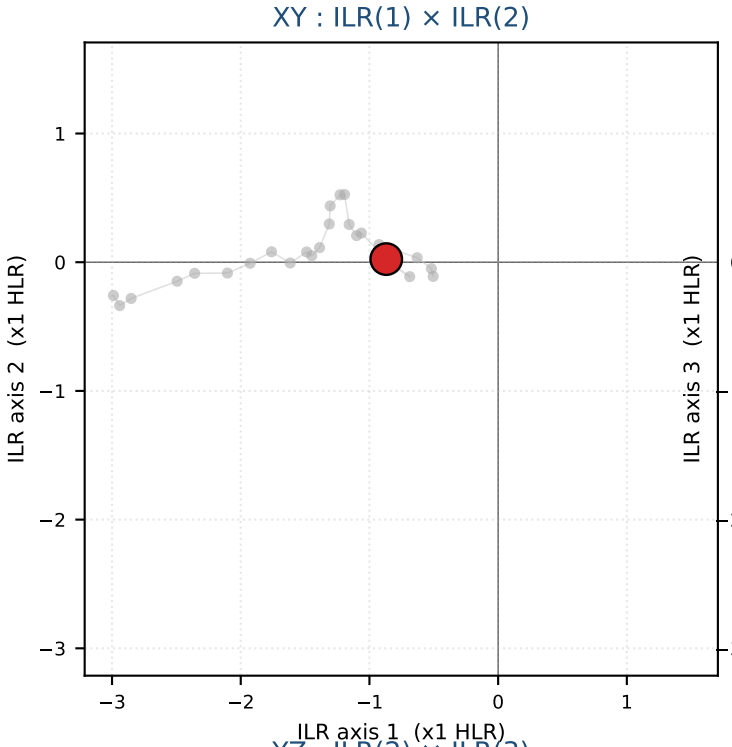
display factor: x1 (raw HLR x 1)

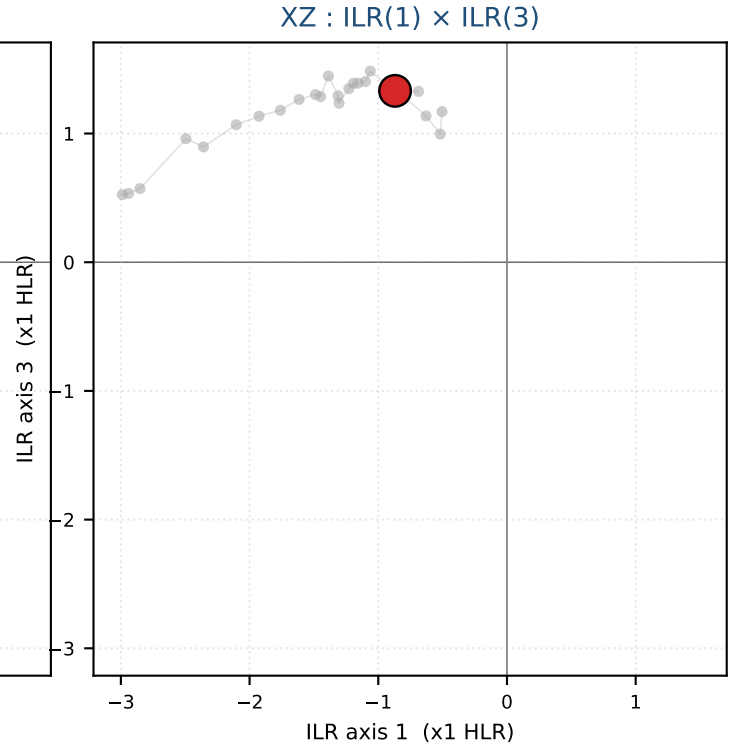
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=19 | 2019 | D=9 T=26 | level 1 first-principles

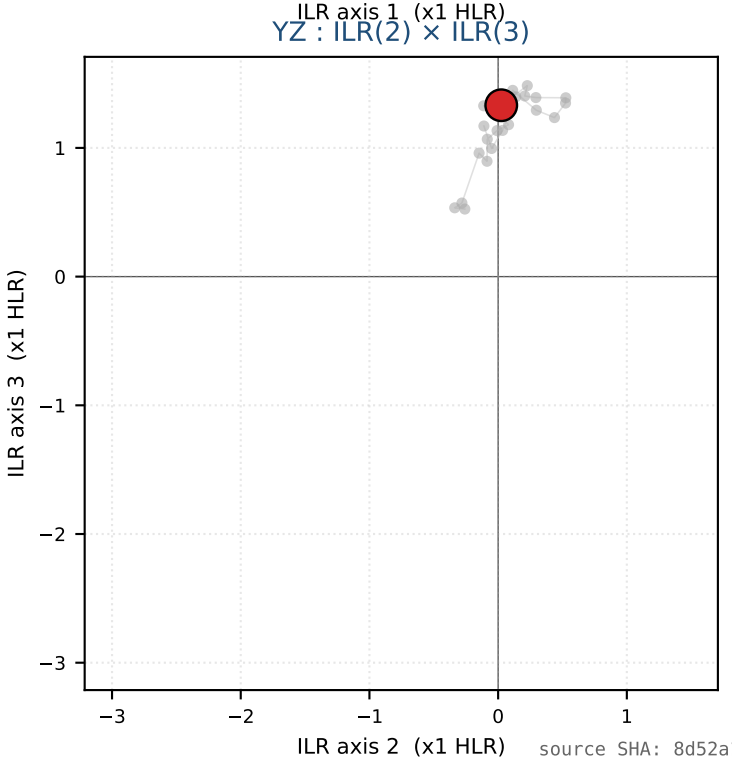
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



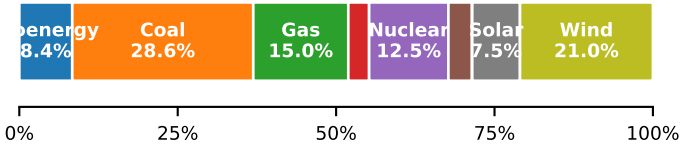


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0835	+0.470
Coal	0.2857	+1.699
Gas	0.1500	+1.055
Hydro	0.0329	-0.463
Nuclear	0.1251	+0.873
Other Fossil	0.0373	-0.336
Other Renewabl	0.0003	-5.055
Solar	0.0754	+0.366
Wind	0.2098	+1.390



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

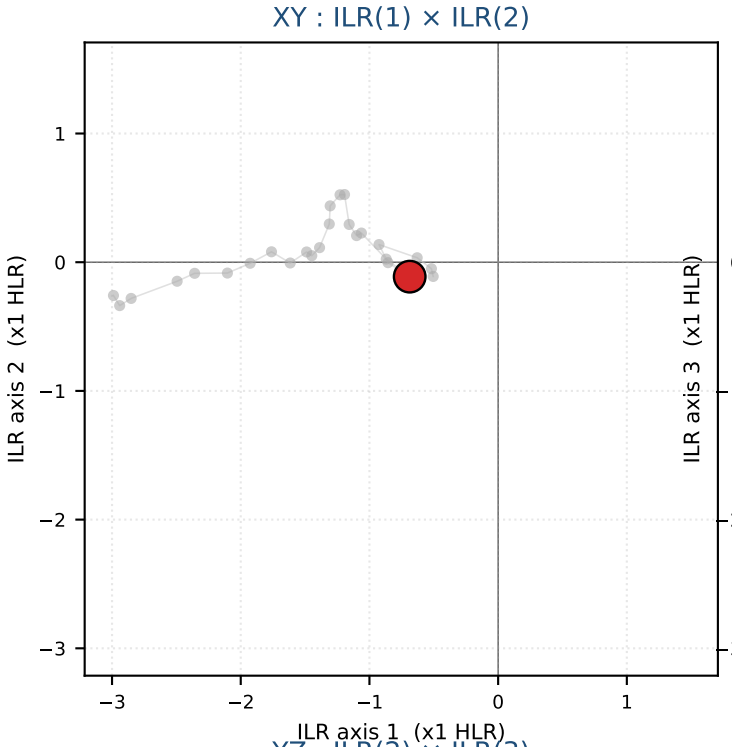
display factor: x1 (raw HLR x 1)

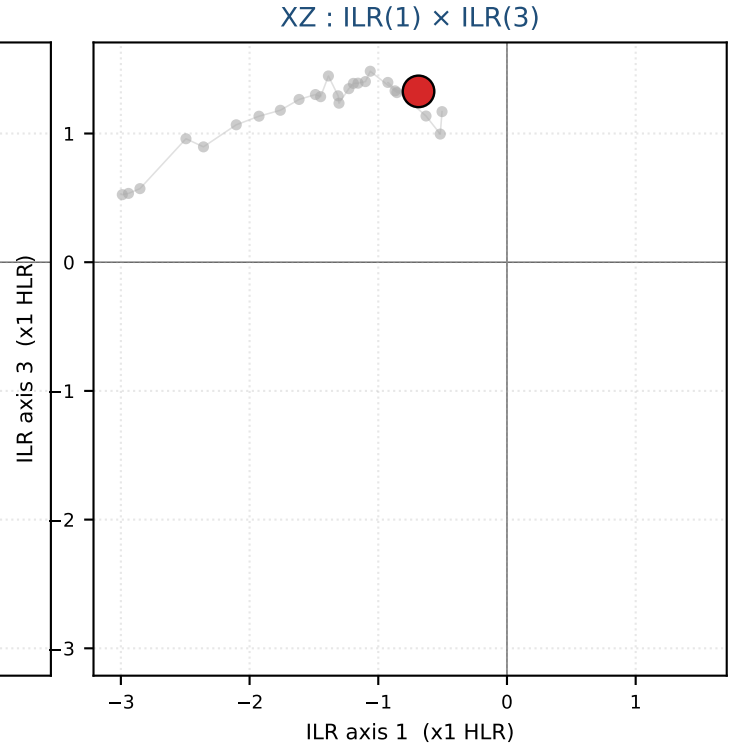
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=20 | 2020 | D=9 T=26 | level 1 first-principles

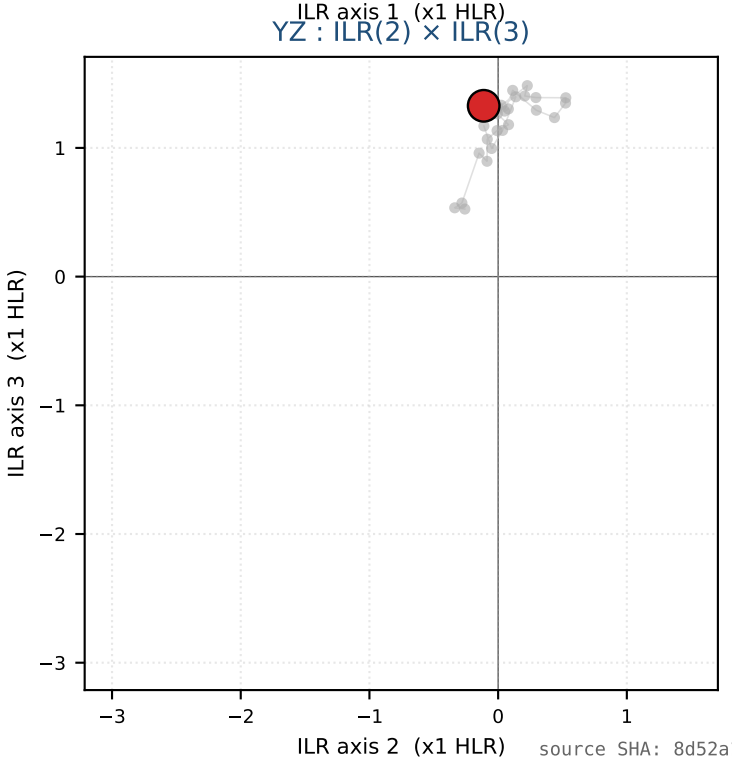
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



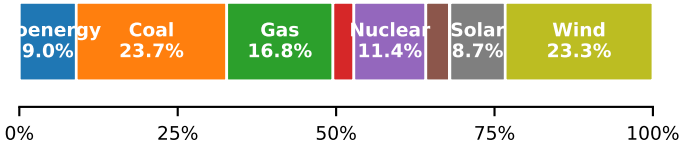


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0899	+0.502
Coal	0.2375	+1.474
Gas	0.1676	+1.126
Hydro	0.0330	-0.499
Nuclear	0.1136	+0.736
Other Fossil	0.0375	-0.371
Other Renewabl	0.0004	-4.898
Solar	0.0873	+0.474
Wind	0.2331	+1.455



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

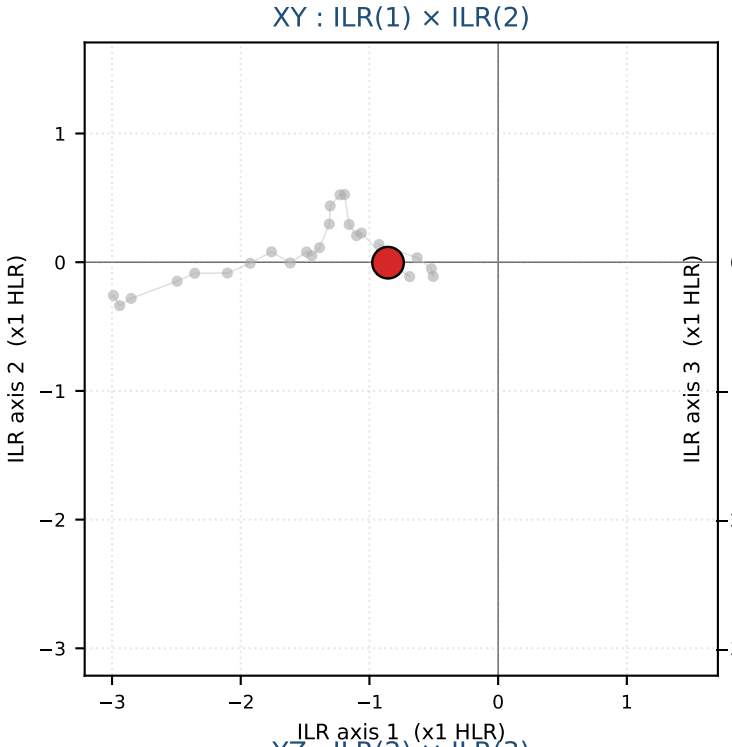
display factor: x1 (raw HLR x 1)

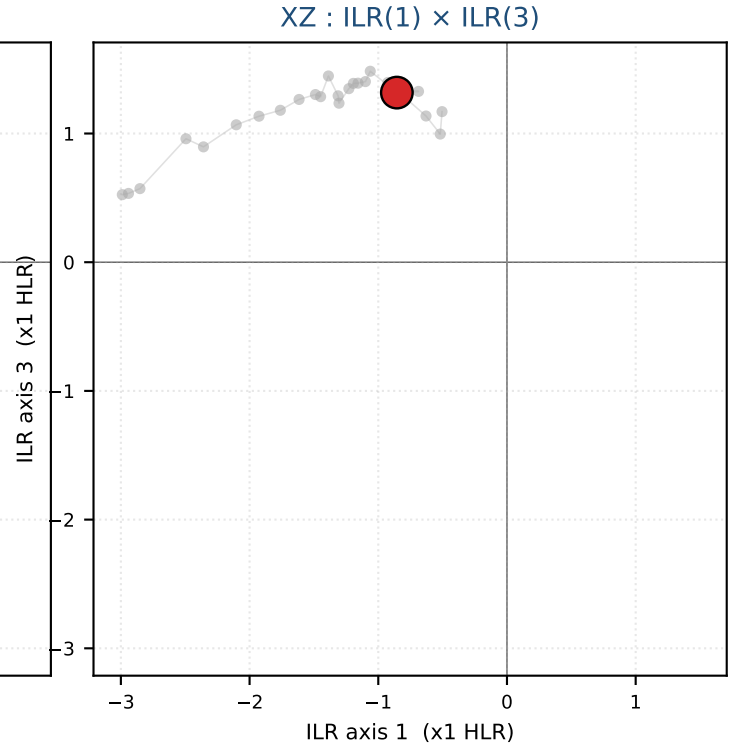
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=21 | 2021 | D=9 T=26 | level 1 first-principles

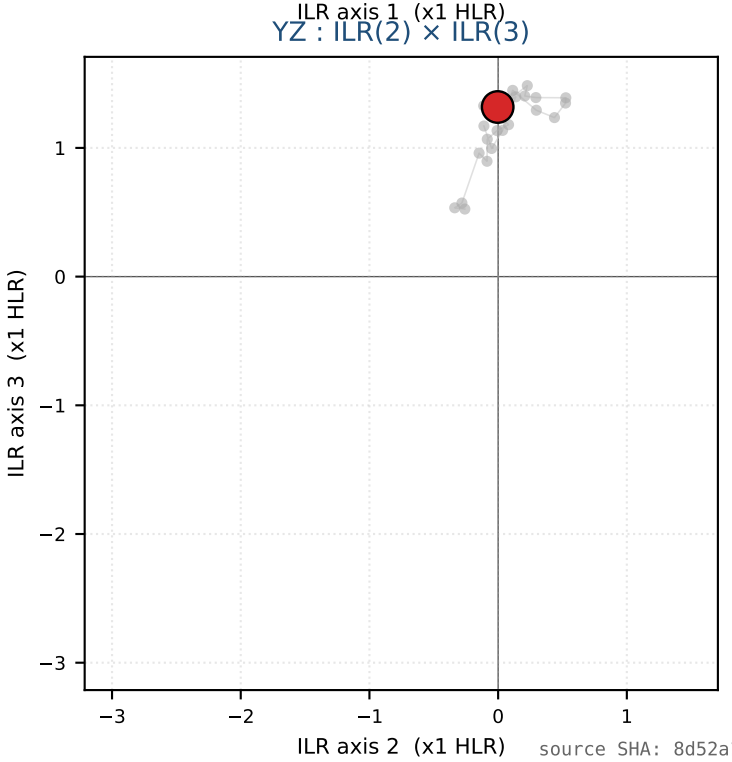
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

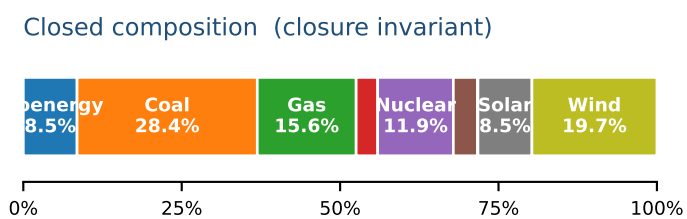




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0848	+0.446
Coal	0.2845	+1.656
Gas	0.1560	+1.056
Hydro	0.0340	-0.469
Nuclear	0.1194	+0.789
Other Fossil	0.0384	-0.346
Other Renewabl	0.0004	-4.875
Solar	0.0852	+0.451
Wind	0.1972	+1.290





closure: Σ = 1.000000 OK

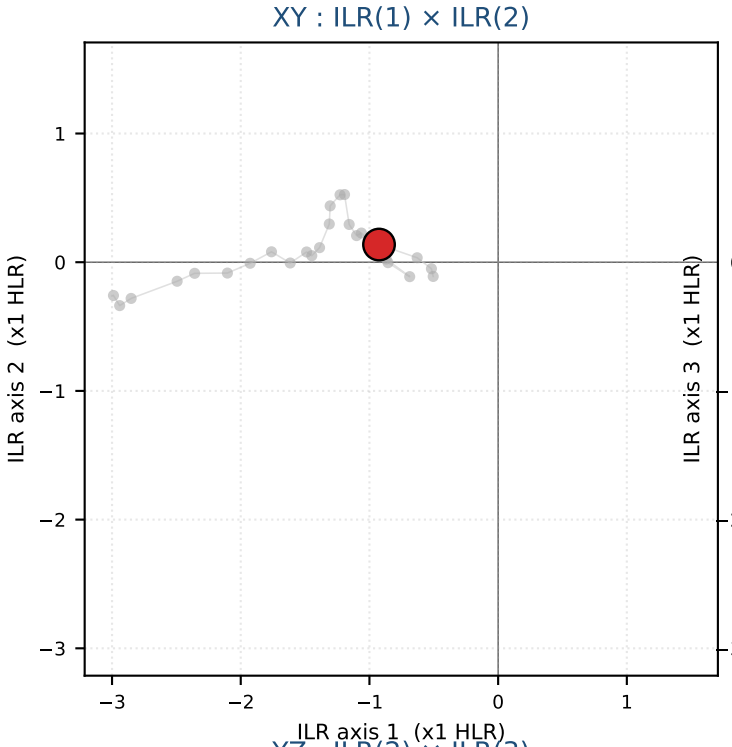
display factor: x1 (raw HLR x 1)

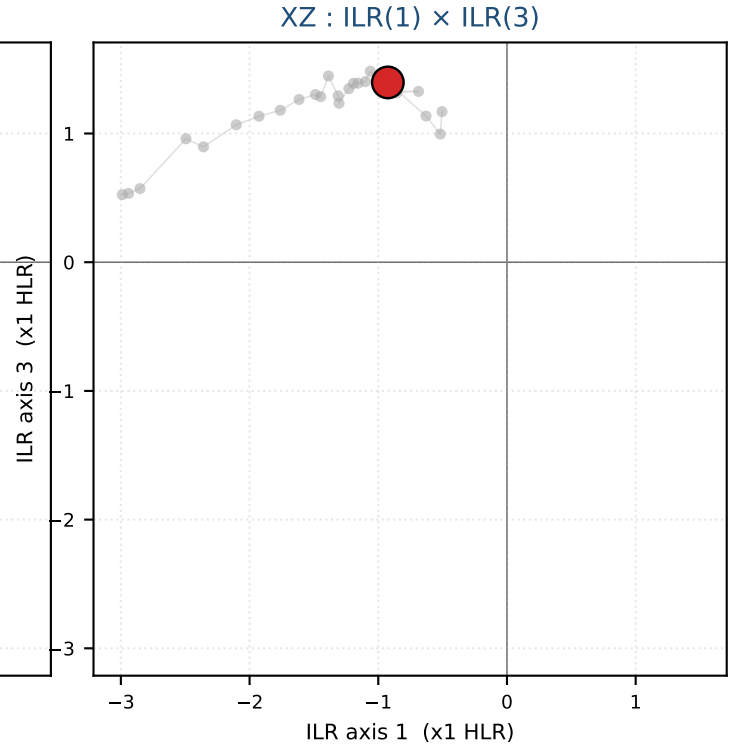
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=22 | 2022 | D=9 T=26 | level 1 first-principles

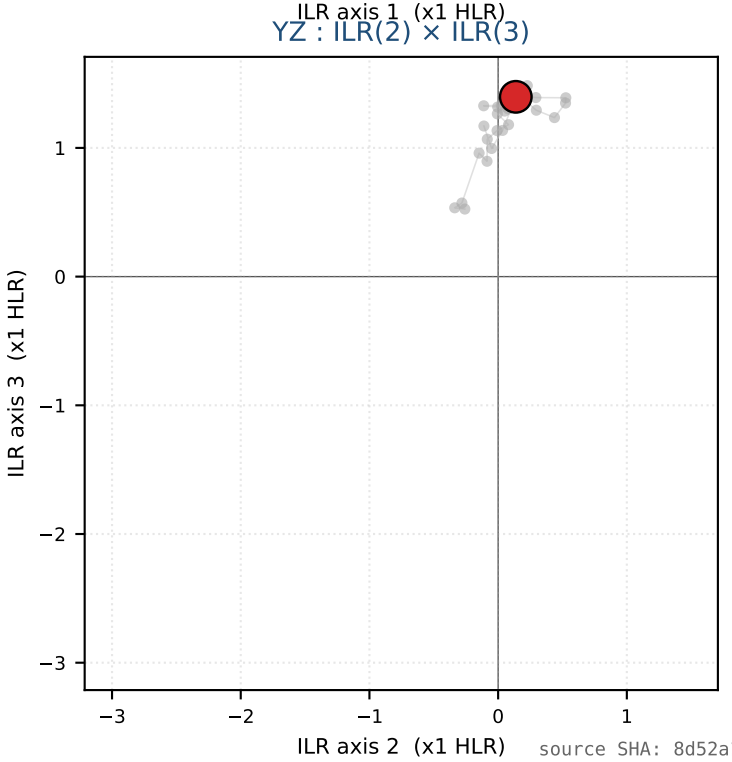
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

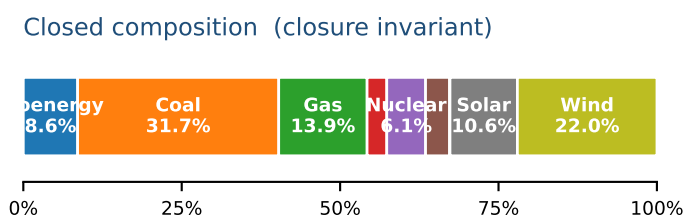




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0857	+0.517
Coal	0.3174	+1.826
Gas	0.1394	+1.003
Hydro	0.0311	-0.498
Nuclear	0.0612	+0.180
Other Fossil	0.0382	-0.292
Other Renewabl	0.0004	-4.928
Solar	0.1064	+0.732
Wind	0.2202	+1.460





closure: Σ = 1.000000 OK

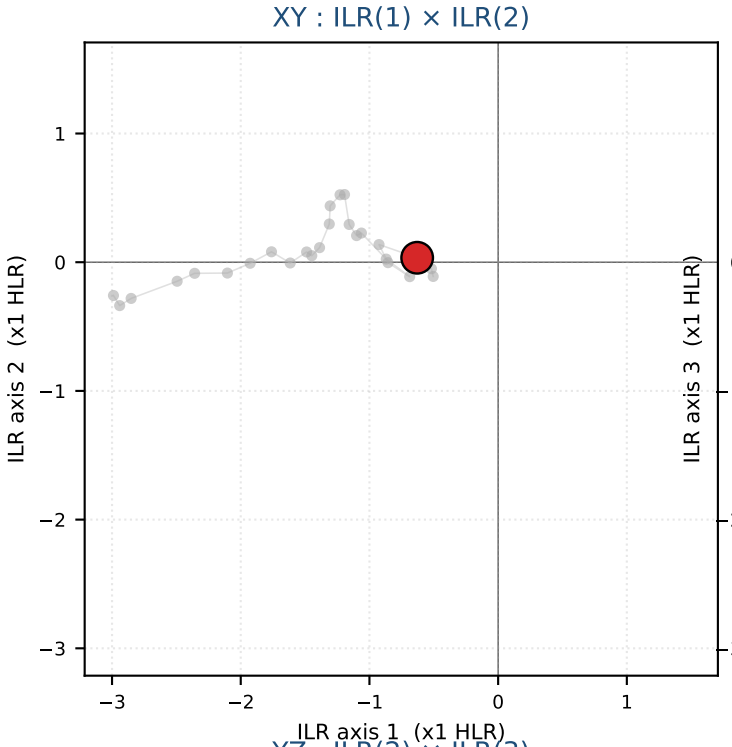
display factor: x1 (raw HLR x 1)

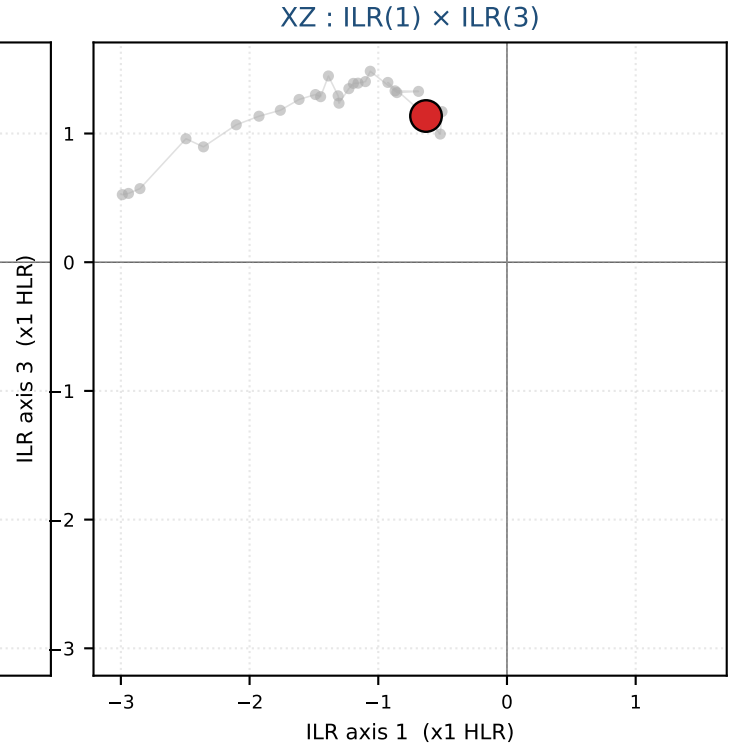
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=23 | 2023 | D=9 T=26 | level 1 first-principles

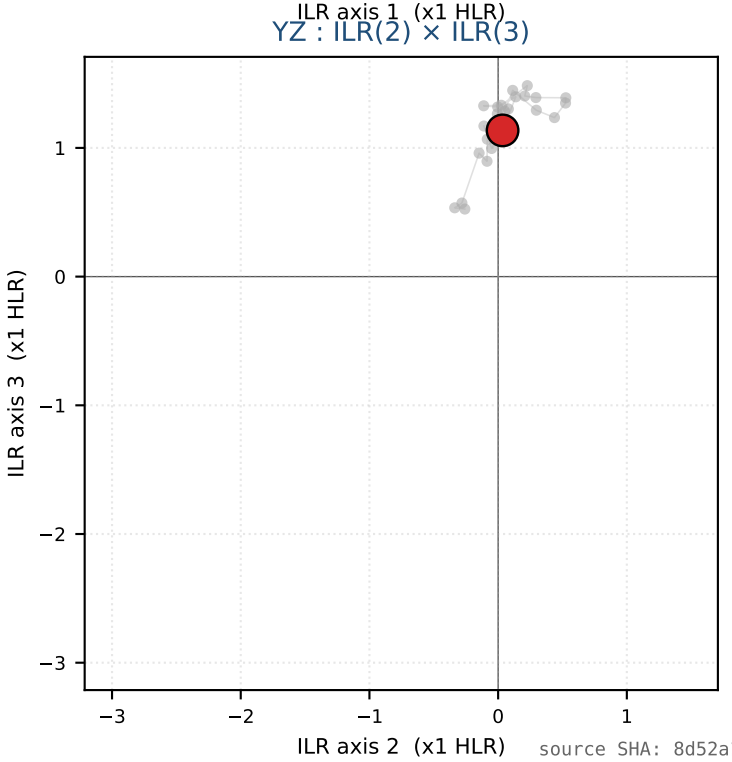
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



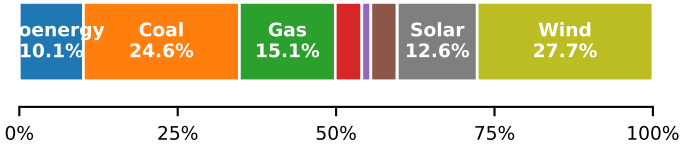


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1012	+0.757
Coal	0.2463	+1.647
Gas	0.1513	+1.159
Hydro	0.0419	-0.124
Nuclear	0.0142	-1.203
Other Fossil	0.0414	-0.137
Other Renewabl	0.0004	-4.841
Solar	0.1260	+0.977
Wind	0.2774	+1.766



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

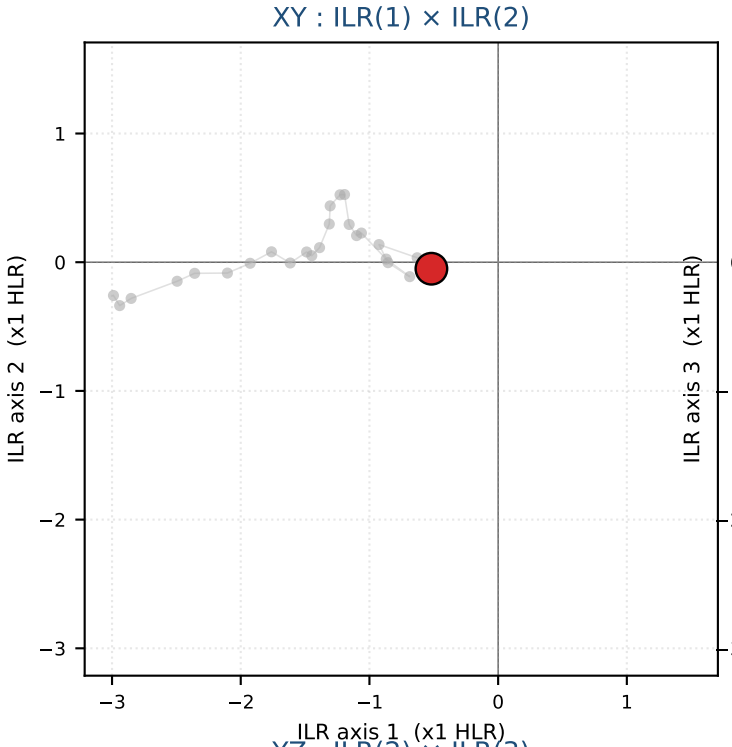
display factor: x1 (raw HLR x 1)

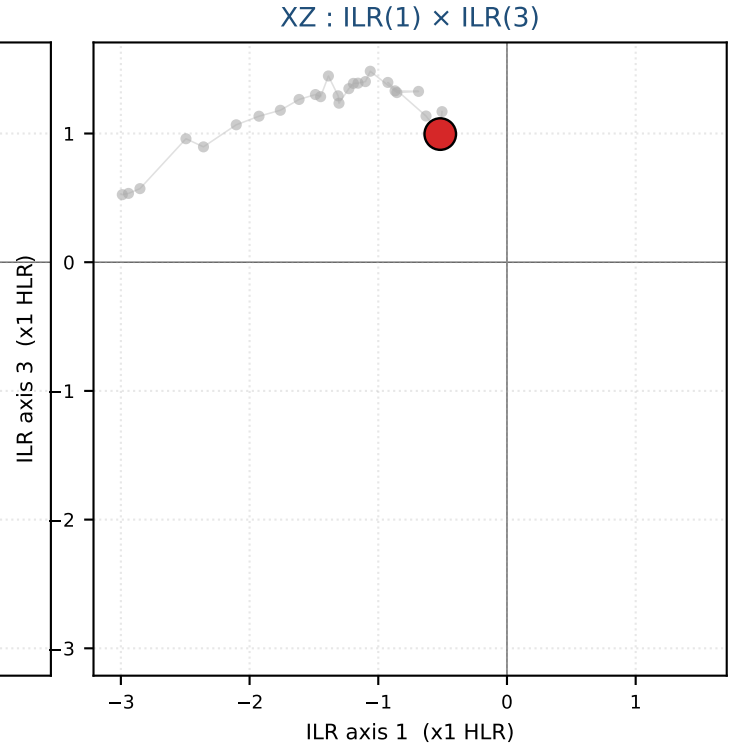
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=24 | 2024 | D=9 T=26 | level 1 first-principles

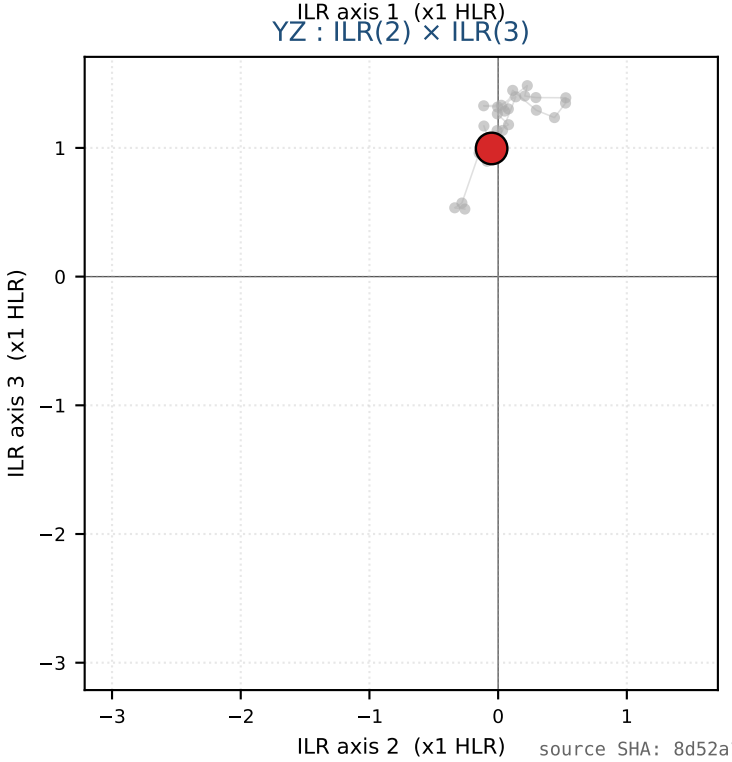
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

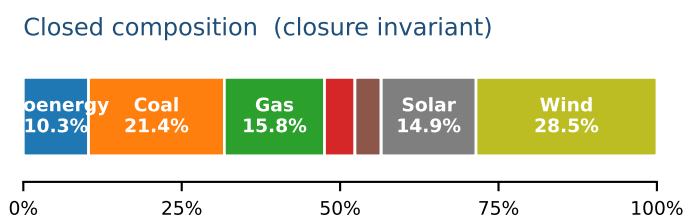




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1030	+4.783
Coal	0.2144	+5.516
Gas	0.1581	+5.212
Hydro	0.0480	+4.021
Nuclear	0.0000	-33.689
Other Fossil	0.0411	+3.865
Other Renewabl	0.0004	-0.665
Solar	0.1495	+5.155
Wind	0.2855	+5.803





closure: Σ = 1.000000 OK

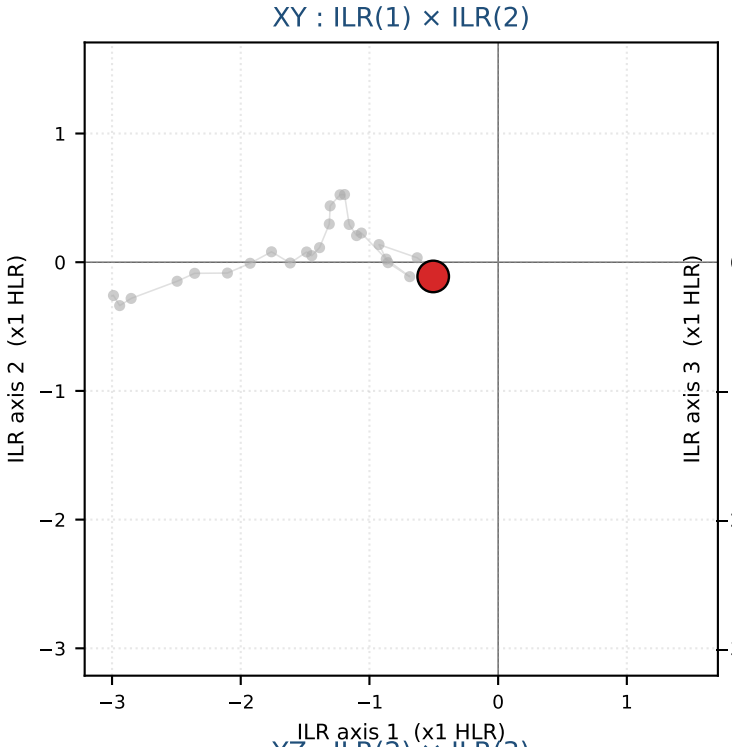
display factor: x1 (raw HLR x 1)

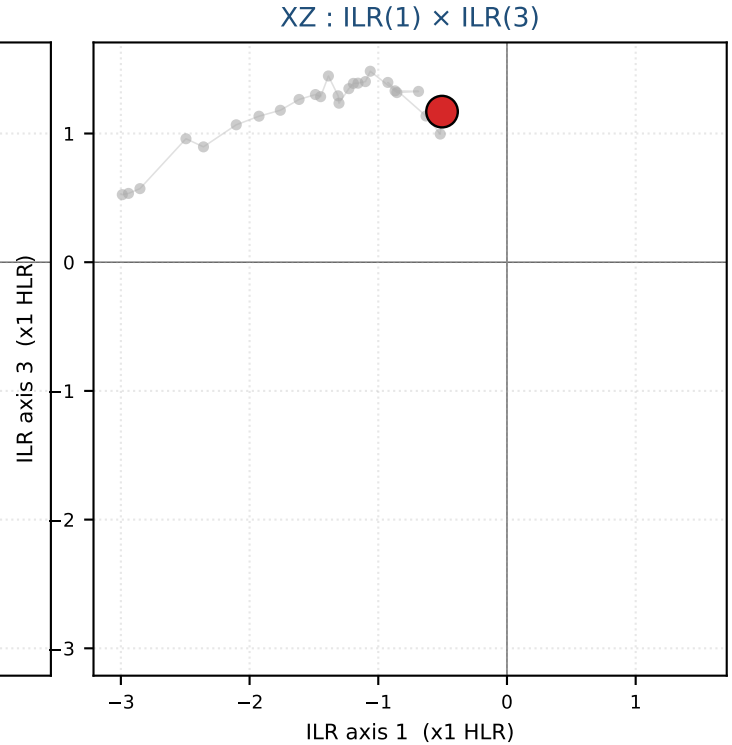
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=25 | 2025 | D=9 T=26 | level 1 first-principles

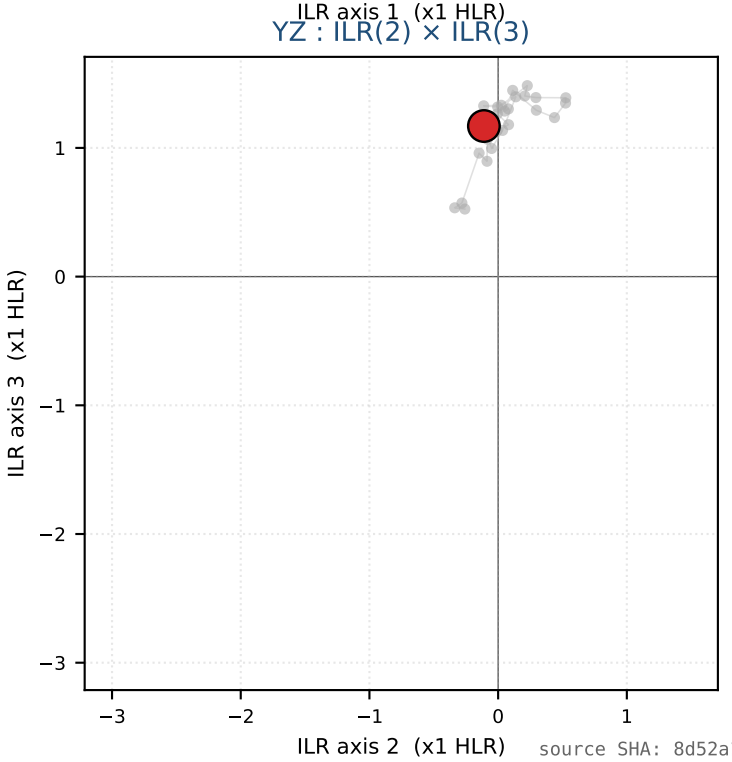
ember_deu | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

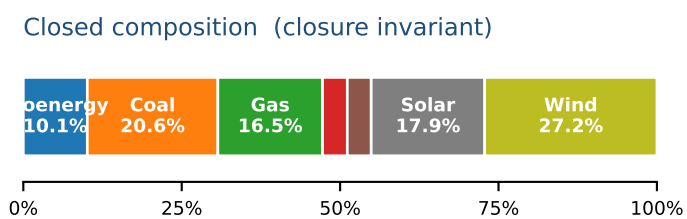




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1010	+8.454
Coal	0.2061	+9.168
Gas	0.1652	+8.946
Hydro	0.0391	+7.505
Nuclear	0.0000	-30.007
Other Fossil	0.0378	+7.471
Other Renewabl	0.0000	-30.007
Solar	0.1791	+9.027
Wind	0.2718	+9.444





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

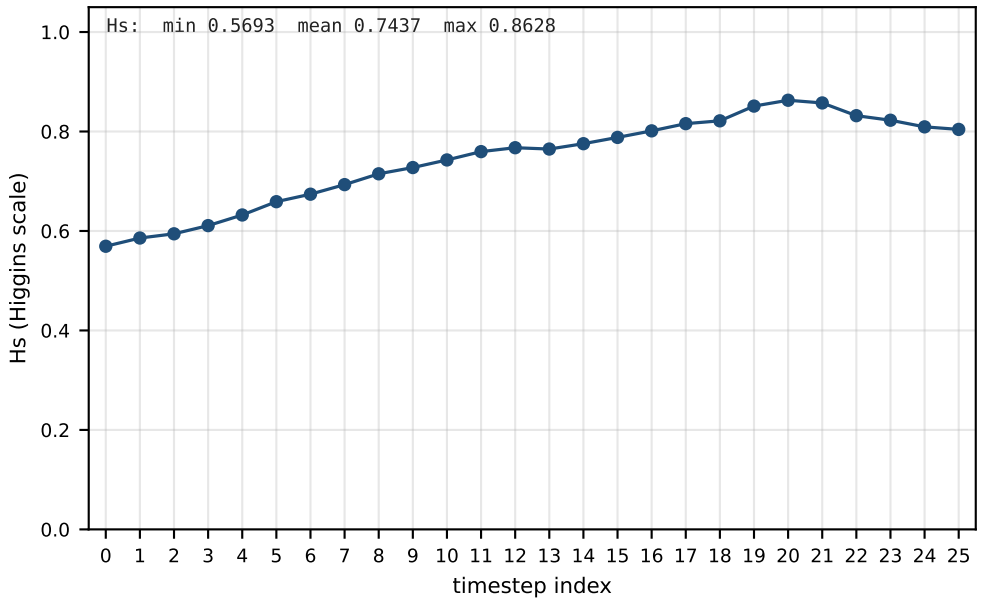
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

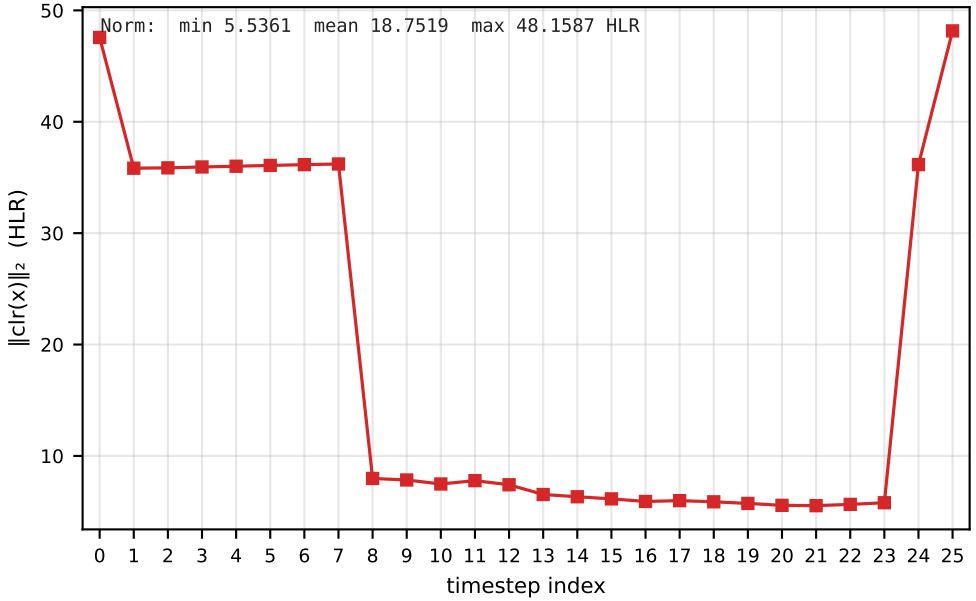
STAGE 1 SUMMARY | ember_deu | D=9 T=26 | Order 1 (first-principles closure)

Engine cnt 2.0.4 | Schema 2.1.0 | Generated 2026-05-06T01:24:14

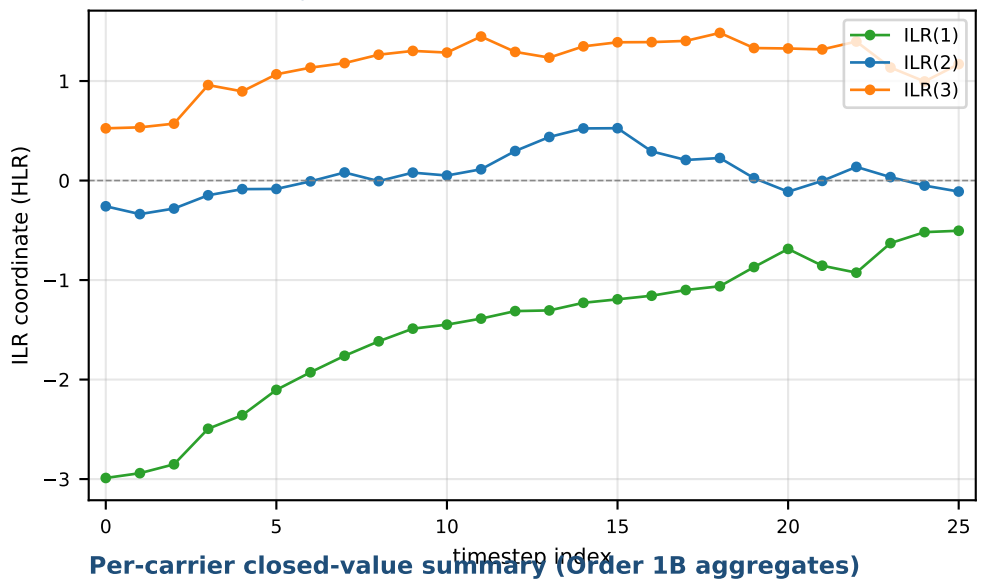
Hs trajectory — single-composition entropy / ln(D)



Aitchison norm — distance from centroid (HLR)



ILR axes 1–3 trajectories (Helmert basis)



Order-1 invariants (whole trajectory)

```
closure invariant     :  max |Σ closed − 1| = 2.22e-16
closure status        :  OK ✓

Hs range              :  [0.5693, 0.8628]
Aitchison norm range  :  [5.5361, 48.1587] HLR
ILR(1) range          :  [-2.9890, -0.5046]
ILR(2) range          :  [-0.3373, +0.5258]
ILR(3) range          :  [+0.5248, +1.4838]

Source file           :  ember_DEU_Germany_generation_TWh.csv
Source SHA-256        :  8d52a1c20cd36340bef837925ccaf037...
Closed-data SHA-256   :  fe6c6f3cbe1098771910d0e270ac4b0f...
Content SHA-256       :  2c9471359a4e0cc328b334d36ccd4013...
Ordering              :  temporal=True, method=by-time
Zero replacement δ    :  1e-15
```

Per-carrier closed-value summary (Order 1B aggregates)

#	carrier	min	mean	max	std
0	Bioenergy	0.0076	0.0596	0.1030	0.0309
1	Coal	0.2061	0.3987	0.5215	0.0963
2	Gas	0.0865	0.1271	0.1676	0.0232
3	Hydro	0.0280	0.0343	0.0480	0.0044
4	Nuclear	0.0000	0.1718	0.2981	0.0894
5	Other Fossil	0.0317	0.0390	0.0433	0.0028
6	Other Renewabl	0.0000	0.0001	0.0004	0.0002
7	Solar	0.0000	0.0496	0.1791	0.0492
8	Wind	0.0164	0.1199	0.2855	0.0850